New GAPN Functions with the Lowest Algebraic Degree

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Abstract

Most of the known (Almost) Perfect Nonlinear functions over an arbitrary finite field $\mathbb{F}_{p^n}$ have the lowest possible algebraic degree (2) and can be represented by Dembowski-Ostrom (DO) polynomials. We define Generalized DO (GDO) polynomials as homogeneous polynomials of algebraic degree p over finite fields with characteristic p. We illustrate the similarities between DO and GDO polynomials when studied through generalized derivatives. We also discuss the differential uniformity and Walsh coefficients of GDO functions. Our main contribution is to use this framework in order to propose several new classes of Generalized Almost Perfect Nonlinear (GAPN) functions of algebraic degree p, the lowest possible over the finite field $\mathbb{F}_{p^n}$.

1 Introduction

(Almost) Perfect Nonlinear (APN/PN) functions have the best resistance against differential cryptanalysis [22] of block ciphers. Due to their properties, APN/PN functions have applications in other domains such as finite geometry [10] and coding theory [35]. As such exceptional objects, APN/PN functions have attracted the attention of many cryptographers and mathematicians over the years (see [25, 33, 15] to name only a few).

In recent years, significant advances have been made in the fields of zero-knowledge protocols [3] and fully homomorphic encryption [8], for instance. These recent developments have moved the paradigm from symmetric cryptographic primitives over binary fields to what is now called arithmetization-oriented cryptography, that is, cryptographic primitives operating on elements belonging to finite fields with odd characteristics.

Among the many attempts to generalize the concept of Almost Perfect Nonlinearity [1, 5, 12, 6], the one introduced by Kuroda: Generalized Almost Perfect Nonlinear (GAPN) functions [17] fits particularly well in odd characteristic as a direct transposition from APN functions in even characteristic. In this article, we provide a brief overview of the known GAPN functions from the literature [17, 16, 24, 27, 2, 36, 37, 30], that we classify in various tables. We display known monomial and non-monomial GAPN functions of algebraic degree p in Table 1 and Table 2 respectively, including our results. In Appendix A, Table 3 and Table 4 present the known monomial and non-monomial GAPN functions that have an algebraic degree strictly greater than p.

Our main contribution is the introduction of new classes of GAPN functions of algebraic degree p, including both monomial and multinomial (see Table 1 and Table 2). For most of these newly introduced GAPN functions, it is possible to check that they are new up to Generalized Extended Affine Equivalence (see Definition 9) because their GAPN conditions differ from previously known ones, for example, some functions are GAPN over some extensions, while others are not. Unlike

most existing constructions, our GAPN monomials support flexible selection of the Hamming degree (see Definition 2).

Exponents e	**Conditions**	**Hamming degree**	**References**
$1+p^{i_2}+\cdots+p^{i_p}$	$i_2,\ldots,i_p>0$, $\{z\in\mathbb{F}_{p^n} \mid z+z^{p^{i_2}}+\cdots+z^{p^{i_p}}=0\}=\mathbb{F}_p$	p	[17, Lemma 3.3]
$k_2p^{l_2}+k_1p^{l_1}$	$l_1<l_2,\ 0<k_1,k_2\leq p-1$, $k_1+k_2=p$, $\gcd(l_2-l_1,n)=1$, $\gcd(k_1+k_2-(p-1),p^n-1)=1$	2	[27, Theorem 2]
$\sum_{i=0}^{\ell}k_i.p^i$	$0\leq k_i\leq p-1$, $n=q_1^{i_1}\cdots q_s^{i_s}\geq 3$, $q_i\notin\{2,p\}$ primes, $\mathbb{O}_{q_i}(p)\geq\ell$ for every $1\leq i\leq s$.	$\in[2,\ell+1]$	Theorem 15
$\sum_{i=0}^{\ell}k_i.p^i$,	$0\leq k_i\leq p-1$, $n=2N$, $N=q_2^{i_2}\cdots q_s^{i_s}\geq 3$, $q_i\notin\{2,p\}$ primes, $\mathbb{O}_{q_i}(p)\geq\ell$ for every $2\leq i\leq s$, $\exists\, 0\leq i,j\leq\ell,\ k_{2i}\neq 0$ and $k_{2j+1}\neq 0$	$\in[2,\ell+1]$	Theorem 18
$k_2p^2+k_1p+k_0$	$n=p^{\alpha}.N$ where $N\not\equiv 0 \pmod p$ and $\alpha\geq 0$ $k_0^N\neq k_2^N$ or ($k_0=k_1$ and $\alpha=0$)	$\in[2,3]$	Theorem 19

Table 1: Known GDO GAPN monomial functions over $\mathbb{F}_{p^n}$ for an odd prime p up to GEA-equivalence.

The article is structured as follows. Section 2 introduces the mandatory notions concerning the different degrees of functions over a finite field and the derivatives of a function. In Section 3, we recall the concept of generalized derivatives, some basic properties, and the so-called Generalized Extended-Affine (GEA) equivalence. Section 4 discusses a generalization of Dembowski-Ostrom (DO) polynomials, that we call generalized DO (GDO) polynomials, that is, homogeneous polynomials of algebraic degree p, the field's characteristic. We emphasize the relevance of considering GDO polynomials for the study of GAPN functions, because GDO polynomials possess the minimal possible algebraic degree to be GAPN. In this section, we also present some cryptographic properties of GDO functions, namely their differential uniformity and their Walsh coefficients. Section 5 presents new classes of GAPN functions of the lowest possible algebraic degree p, i.e., GAPN GDO functions. First, we treat the case of monomial GAPN functions. Then, because generalized derivatives of GDO functions can be expressed as linearized polynomials, we present new multinomial GAPN functions and, more notably, homogeneous GAPN GDO functions of Hamming degree 3 and 5.

Polynomial Representation	Conditions	Hamming degree	References
$x^{p^i+p-1} - x^{p^{n-i}+p-1}$	$i > 0$ and $gcd(i,n) = 1$	2	[17, Proposition 3.5]
$\sum_{i=1}^{p-1} c^{p-1-i} x^{ip^{l_2}+(p-i)p^{l_1}}$	$l_2 > l_1,\ gcd(l_2 - l_1, n) = 1$ $c = \alpha^v,\ \alpha$ primitive element of $\mathbb{F}_{p^n}$ $v \mid p-1$ and $v < p-1$	2	[27, Theorem 7]
$cx^{i_1 p^{l_2}+(p-i_1)p^{l_1}} + x^{i_2 p^{l_2}+(p-i_2)p^{l_1}}$	$l_2 > l_1,\ gcd(l_2 - l_1, n) = 1,$ $1 \le i_1 < i_2 \le p-1,$ $c = (-1)^{i_2-i_1+1} \frac{\binom{p-2}{i_2-1}}{\binom{p-2}{i_1-1}} \beta,$ $\beta \in \mathbb{F}_{p^n}$ of order not dividing $\frac{p^n-1}{i_3(p-1)}$ $i_3 = gcd(i_2 - i_1, \frac{p^n-1}{p-1}) = 1$	2	[27, Theorem 8]
$x^{p-s}(ax^{p^k} + x^{p^{n-k}} + bx)^s$	$n \ge 3,\ 1 \le k \le n-1,$ $gcd(n,k) = 1$ $2 \le s \le p-1,$ $a^{\frac{p^n-1}{p-1}} \ne 1,$ $b \notin \{0, \frac{ax^{p^k}+x^{p^{n-k}}}{x} \mid x \in \mathbb{F}_{p^n}^*\}$	$\in [2, s+1]$	[27, Example 3]
$x^{p-s}(ax^{p^k} + x^{p^{n-k}} + bx)^s$	$n \ge 3,\ 1 \le k \le n-1,$ $gcd(n,k) = 1$ $2 \le s \le p-1,$ $a^{\frac{p^n-1}{p-1}} = 1,\ n$ odd $b \notin \{0, \frac{ax^{p^k}+x^{p^{n-k}}}{x} \mid x \in \mathbb{F}_{p^n}^*\}$	$\in [2, s+1]$	[27, Example 3]
$x^{p-s}(ax^{p^k} + x^{p^{n-k}})^s$	$n \ge 3$ odd, $1 \le k \le n-1,$ $gcd(n,k) = 1$ $2 \le s \le p-1,$ $a^{\frac{p^n-1}{p-1}} \notin \{-1, 1\},$	$\in [2, s+1]$	[27, Corollary 3]
$x^{d_1} + ux^{d_2}$ (over $\mathbb{F}_{p^2}$)	$d_1, d_2 \in [1, p^2-1],\ d_2$odd and u not a square in$\mathbb{F}_{p^2}$ $x \mapsto x^{d_1}$ GAPN over $\mathbb{F}_{p^2}$	2	[2, Theorem 1]
$x^{d_1} + ux^{d_2}$ (over $\mathbb{F}_{p^2}$)	$d_1, d_2 \in [1, p^2-1],\ d_2$ even and $\exists\, N$ odd such that $N \mid p+1$ and $N \mid d_2 - d_1$ and u not an N-th power in $\mathbb{F}_{p^2}$ $x \mapsto x^{d_1}$ GAPN over $\mathbb{F}_{p^2}$	2	[2, Theorem 1]
$x^{2p-1} + ux^{kp+l}$ (over $\mathbb{F}_{p^2}$)	$u \in \mathbb{F}_{p^2}$ a primitive element, $k, l \in [0, p-1]$ with $l + k$ odd	2	[2, Corollary 1]
$x^{(p-d_0)p^2+d_0 p} + \lambda x^{k_0 p+p-k_0}$	$(\frac{k_0}{d_0})^n \lambda^{\frac{p^n-1}{p-1}} \ne 1$	2	Corollary 5
$x^{e_3(i)} - \lambda x^{e_3(j)}$	$p > 3,\ e_3(k)$ as in Definition 15 $\frac{i}{j}\lambda$ not an $e_3(i) - e_3(j)$ power in $\mathbb{F}_{p^n}^*,\ n \ge 3.$	3	Corollary 6
$x^{e_5(i)} - \lambda x^{e_5(j)}$	$p \ge 17$ such that $p \ne 19,$ $e_5(k)$ as in Definition 16, $\lceil \frac{p+1}{6} \rceil \le i \ne j \le \lfloor \frac{p-1}{4} \rfloor,$ $\frac{i}{j}\lambda$ not an $e_5(i) - e_5(j)$ power in $\mathbb{F}_{p^n}^*,\ n \ge 5.$	5	Corollary 9
$x^{e_3(1)} + \frac{f}{2} x^{e_3(2)} + \frac{f^2-\lambda}{12} x^{e_3(3)}$	$p \ge 7,\ n \ge 3$ such that $n \not\equiv 0 \pmod p,$ $e_3(k)$ as in Definition 15, $f \in \mathbb{F}_{p^n},$ $\lambda \in \mathbb{F}_{p^n}$ not a square.	3	Corollary 8
$\frac{1}{i} x^{e_5(i)} + \frac{f}{j} x^{e_5(j)} + \frac{f^2-\lambda}{4k} x^{e_5(k)}$	$p \ge 17$ such that $p \ne 19,\ n \ge 5$ such that $n \not\equiv 0 \pmod p,$ $e_5(k)$ as in Definition 16, $i = \lceil \frac{p+1}{6} \rceil,\ j = i+1,\ k = i+2$ $f \in \mathbb{F}_{p^n},$ $\lambda \in \mathbb{F}_{p^n}$ not a square.	5	Corollary 9

Table 2: Known non-monomial GAPN functions which are GDO polynomials over $\mathbb{F}_{p^n}$ for an odd prime p.

2 Preliminaries

We denote by $\mathbb{F}_q$ the finite fields with q elements, where q is a power of a prime p called the characteristic, and the multiplicative group of its non-zero elements by $\mathbb{F}_q^*$. In this work, we will be using different notions for the degree of a function $F(x) = \sum_{i=0}^{p^n-1} c_i x^i$ over $\mathbb{F}_{p^n}$.

Definition 1 (Polynomial Degree)**.** The polynomial degree of F is defined by:

$$\deg_P(F) := \max\{i \mid 0 \leq i < p^n, \quad c_i \neq 0\}.$$

Definition 2 (Hamming Degree)**.** The Hamming degree of F is defined by:

$$\deg_H(F) := \max\{H_p(i) \mid 0 \leq i < p^n, \quad c_i \neq 0\},$$

where $H_p(i)$ is the p-ary hamming weight of the integer i, that is, the number of non-zero coefficients in the p-ary representation of i.

Definition 3 (Algebraic Degree)**.** The algebraic degree of F is defined by:

$$\deg_A(F) := \max\left\{\sum_{i=0}^{n-1} a_i \mid 0 \leq a_i < p, \quad c_{\sum_{i=0}^{n-1} a_i p^i} \neq 0\right\}.$$

Remark 1. In binary fields, the concepts of Hamming degree and algebraic degree coincide.

Example 1. Let $p = 5$ and $n > 2$ be an integer. The monomial function $x \mapsto x^{40} = x^{1\times 25+3\times 5}$ over $\mathbb{F}_{5^n}$ has:

- polynomial degree, $\deg_P(x^{40}) = 40$,
- Hamming degree, $\deg_H(x^{40}) = 2$,
- Algebraic degree, $\deg_A(x^{40}) = 4$.

Although we do not restate the correspondence here, it is worth noting that the algebraic degree of a univariate function over $\mathbb{F}_{p^n}$ corresponds to the maximum multivariate degree of its component functions when seen over the vector space $\mathbb{F}_p^n$.

Definition 4 (Derivative functions)**.** Let $F : \mathbb{F}_q \to \mathbb{F}_q$ be a function, and let $\alpha \in \mathbb{F}_q^*$ be a non-zero element. The derivative of F in the direction α is defined as:

$$\Delta_\alpha F(x) = F(x + \alpha) - F(x).$$

The derivative of order d along the directions $\alpha_i \in \mathbb{F}_q^*$, for $i = 1, \ldots, d$ is defined by:

$$\Delta_{\alpha_1,\ldots,\alpha_d} F(x) = \Delta_{\alpha_1}\Delta_{\alpha_2,\ldots,\alpha_d} F(x) = \sum_{j=0}^{d} (-1)^{d-j} \sum_{\{i_1,\ldots,i_j\}\subseteq\{1,\ldots,d\}} F(x + \sum_{k=1}^{j} \alpha_{i_k}).$$

We will denote $\Delta_S F(x) := \Delta_{\alpha_1,\ldots,\alpha_d} F(x)$, where $S = \{\alpha_1, \ldots, \alpha_d\}$.

The following proposition is an immediate consequence of [18, Proposition 2].

Proposition 1. *With the same notation as in Definition 4,*

$$\deg_A(\Delta_S F) \leq \deg_A(F) - \#S,$$

where $\#S$ is the cardinality of the set S.

Since our study also relies on linearized polynomials, we now introduce some related notions. See [19, Chapter 3] for more details.

Definition 5 (Definition 3.49 [19])**.** Let q be a prime power. A polynomial of the form

$$L(x) = \sum_{i=0}^{n} a_i x^{q^i} \in \mathbb{F}_{q^m}[x]$$

with coefficients in an extension field $\mathbb{F}_{q^m}$ is called q-polynomial over $\mathbb{F}_{q^m}$ (or, equivalently, q-linearized polynomial over $\mathbb{F}_{q^m}$).

In the following, we refer to p-polynomials as just linearized polynomials.

Definition 6 (Definition 3.58 [19])**.** Let q be a prime power. The polynomials

$$l(x) = \sum_{i=0}^{n} a_i x^i \quad \text{and} \quad L(x) = \sum_{i=0}^{n} a_i x^{q^i}$$

over $\mathbb{F}_{q^m}$ are called q-associates of each other. More specifically, $l(x)$ is the conventional q-associate of $L(x)$ and $L(x)$ is the linearized q-associate of $l(x)$.

3 Generalized differentials

In 2017, Kuroda and Tsujie [17] proposed a generalization of APN functions that is particularly well suited to fields of odd characteristic, primarily because of its similarity with APN functions defined over a binary field. More specifically, they defined generalized almost perfect nonlinear (GAPN) functions to be functions $F : \mathbb{F}_{p^n} \to \mathbb{F}_{p^n}$ for which the equations

$$\sum_{i \in \mathbb{F}_p} F(x + i\alpha) = \beta, \qquad \forall\, \alpha \neq 0,\ \beta \in \mathbb{F}_{p^n},$$

have at most p solutions. Actually, this notion can be extended to the concept of generalized differential uniformity.

Definition 7 (Generalized differential uniformity)**.** The generalized differential uniformity of a function $F : \mathbb{F}_{p^n} \to \mathbb{F}_{p^n}$ is the value:

$$\mathcal{U}(F) = \max_{\alpha \neq 0, \beta \in \mathbb{F}_{p^n}} \#\{x \in \mathbb{F}_{p^n} \mid \sum_{i \in \mathbb{F}_p} F(x + i\alpha) = \beta\} \geq p.$$

When $\mathcal{U}(F) = p$, the function F is called a GAPN function.

It follows directly from this definition that the generalized differential uniformity is a multiple of the underlying field characteristic.

Later, Salagean and Ozbudak [24, 27] used discrete derivatives to interpret GAPN properties. We recall their definition.

Definition 8 (Generalized derivative functions)**.** The Generalized derivative of a function $F : \mathbb{F}_{p^n} \to \mathbb{F}_{p^n}$ in a direction $\alpha \neq 0 \in \mathbb{F}_{p^n}$ is defined as:

$$\nabla_\alpha F(x) := \sum_{i \in \mathbb{F}_p} F(x + i\alpha).$$

Proposition 2. *Let $F : \mathbb{F}_{p^n} \to \mathbb{F}_{p^n}$ and $\alpha \in \mathbb{F}_{p^n}$. We have the following identities for all $x \in \mathbb{F}_{p^n}$:*

1. $\nabla_\alpha F(x) = \Delta_{\underbrace{\alpha, \alpha, \ldots, \alpha}_{p-1}} F(x)$,
2. $\nabla_\alpha F(x) = -\Delta_{\alpha, 2\alpha, \ldots, (p-1)\alpha} F(x)$,
3. $\nabla_\alpha F(x) = \sum_{i \in \mathbb{F}_p^*} \Delta_{i\alpha} F(x)$.

Proof. 1. See [24, Lemma 1].

2. See [29] for a detailed proof.

3.

$$\begin{aligned}
\sum_{i \in \mathbb{F}_p^*} \Delta_{i\alpha} F(x) &= \sum_{i \in \mathbb{F}_p^*} (F(x + i\alpha) - F(x)) \\
&= (\sum_{i \in \mathbb{F}_p^*} F(x + i\alpha)) - (p-1)F(x) \\
&= (\sum_{i \in \mathbb{F}_p^*} F(x + i\alpha)) + F(x) \\
&= \sum_{i \in \mathbb{F}_p} F(x + i\alpha) \\
&= \nabla_\alpha F(x).
\end{aligned}$$

□

The third point of Proposition 2 gives us new insights into what a generalized derivative is. To the best of our knowledge, there is no reference to this equality and so we include the proof for the sake of completeness. We can see it as a gradient: the sum of derivatives in all directions $i\alpha$, with $i \in \mathbb{F}_p^*$, of a given function. This justifies the notation ∇ for generalized derivatives.

Similarly to the 'regular' differential operation, the generalized differential operations make the algebraic degree decrease accordingly.

Proposition 3. *Let $F : \mathbb{F}_{p^n} \to \mathbb{F}_{p^n}$ and $\alpha \in \mathbb{F}_{p^n}^*$. Then,*

$$\deg_A(\nabla_\alpha F) \leq \deg_A(F) - (p-1).$$

We recall a result, first established by Xiong et al. [31], that characterizes the set of derivative functions over $\mathbb{F}_q$.

Theorem 4 ([31] Theorem 1)**.** *Let G be a function over $\mathbb{F}_q$. There is a function F such that*

$$G(x) = \Delta_\alpha F(x) = F(x+\alpha) - F(x),$$

for some $\alpha \in \mathbb{F}_q$, if, and only if,

$$\nabla_\alpha G(x) = 0, \forall\, x \in \mathbb{F}_q.$$

In addition to their introduction of generalized derivative functions, Ozubdak and Salagean [24] also generalized the so-called Extended Affine (EA) equivalence into the Generalized Extended Affine (GEA) equivalence.

Definition 9 ([24] Definition 2)**.** Two functions $F_1, F_2 : \mathbb{F}_q \to \mathbb{F}_q$ are said to be generalized extended affine equivalent (GEA-equivalent), if there exist bijective affine functions $A_1, A_2 : \mathbb{F}_q \to \mathbb{F}_q$ and a function $B : \mathbb{F}_q \to \mathbb{F}_q$, where $\deg_A(B) \leq p-1$, such that

$$F_1(x) = A_1 \circ F_2 \circ A_2(x) + B(x).$$

Furthermore, Ozbudak and Salagean [24, Proposition 2] observed that the GAPN property is preserved under GEA-equivalence. The same invariance holds for generalized differential uniformity, as well as for the algebraic degree. We omit the proof of the following proposition, as it is analogous to the proof for EA-equivalence.

Proposition 5. *Let $F_1, F_2 : \mathbb{F}_q \to \mathbb{F}_q$. If the functions F_1 and F_2 are GEA-equivalent then*

- $\mathcal{U}(F_1) = \mathcal{U}(F_2)$.

Moreover, if the functions F_1 and F_2 have an algebraic degree at least $p-1$, then

- $\deg_A(F_1) = \deg_A(F_2)$.

To the best of our knowledge, there is no more general equivalence preserving the generalized differential uniformity. In particular, Kuroda and Tsujie [17] noticed that CCZ-equivalence [7], which corresponds to the affine equivalence of graphs between two functions, does not preserve the generalized differential uniformity. It is well known that CCZ-equivalence does not necessarily preserve the algebraic and Hamming degree, which is one of the reasons why CCZ-equivalence is more general than the EA-equivalence. It is therefore natural to ask the following question.

Open problem 1. Is there an equivalence of functions that preserves the generalized differential uniformity but does not necessarily preserve the algebraic degree?

Remark 2. Generally, GEA-equivalence does not preserve the Hamming degree. For example, let $A_2(x) = x^p + x$ be a p-linearized function over $\mathbb{F}_{p^n}$ and $F(x) = x^{p+p-1}$ be a function over $\mathbb{F}_{p^n}$. We have

$$\begin{aligned} F(A_2(x)) &= (x^p + x)^{p+p-1} = (x^p + x)^p \times (x^p + x)^{p-1} \\ &= (x^{p^2} + x^p) \times \sum_{i=0}^{p-1} \binom{p-1}{i} x^{ip} x^{p-i-1} \\ &= \sum_{i=0}^{p-1} \binom{p-1}{i} \left(x^{p^2+ip+p-i-1} + x^{(i+1)p+p-i-1} \right). \end{aligned}$$

From the expression above, we note that F is GEA-equivalent to $F \circ A_2$ when A_2 is a bijection of $\mathbb{F}_{p^n}$ and although F is of Hamming degree 2, $F \circ A_2$ is of Hamming degree 3.
However, for the GEA-equivalence to preserve the Hamming degree, it is sufficient for the right-hand side bijective affine function, A_2, to be monomial (and that $\deg_A(F) > \deg_A(B)$).

4 GAPN functions of low algebraic degree

Dembowski-Ostrom (DO) polynomials over $\mathbb{F}_{p^n}$ are Hamming Degree 2 polynomials:

$$\sum_{i,j} a_{i,j} x^{p^i+p^j} \in \mathbb{F}_{p^n}[x],$$

which possess numerous algebraic and combinatorial properties (see [13] for a recent development). Most of the (A)PN functions we know, up to EA-equivalence, are represented by DO polynomials. There exist algorithmic methods to construct quadratic APN functions [34, 32, 33], and only one PN function is not of the DO type [9]. Since the algebraic degree of DO polynomials is also 2, their generalized differentials always vanish in odd characteristic, and their study regarding generalized differentials stops there. It is natural to look at the functions of algebraic degree p, the characteristic of the field, which is the lowest algebraic degree for a GAPN function. As expected, a lot of the work regarding the APN properties of DO polynomials in even characteristic extends naturally to the study of the GAPN properties of algebraic degree p functions. In this section, we discuss some of these connections and highlight some notable differences.

Definition 10. We call Generalized Dembowski-Ostrom (GDO) polynomials, homogeneous polynomials of algebraic degree p:

$$F(x) = \sum_{i \in I_p} f_i x^i \in \mathbb{F}_{p^n}[x]$$

where

$$I_p = \left\{ \sum_{j=0}^{n-1} i_j p^j \mid 0 \leq i_j \leq p-1 \ \forall\, 0 \leq j \leq n-1, \ \sum_{j=0}^{p-1} i_j = p \right\}.$$

From Proposition 3, we can easily see that the generalized derivative functions of a GDO polynomial are affine, that is, of algebraic degree at most 1. We can actually explicitly give the coefficients of the generalized derivatives of GDO polynomials. We also notice that the generalized derivative functions of GDO polynomials are linearized polynomials.

Lemma 6. *Let p be an odd prime number and $F(x) = \sum_{i=1}^{N} f_i.x^{d_i}$ where $d_i = \sum_{j=0}^{\ell} d_{i,j} p^j$, be a GDO function over $\mathbb{F}_{p^n}$. Then, for every $\alpha \in \mathbb{F}_{p^n}$,*

$$\nabla_\alpha F(x) = \sum_{j=0}^{\ell} g_j^{(F)}(\alpha) x^{p^j},$$

where $g_j^{(F)}(\alpha) = \sum_{i=1}^{N} -d_{i,j} f_i \alpha^{d_i - p^j}$ for every $j \in [0, \ell]$.

Proof. Let $\alpha \in \mathbb{F}_{p^n}^*$, and for $d \in \mathbb{N}$, let $M_d(x) = x^d$.

Using the linearity of the derivative operator ∇_α, we get:

$$\begin{aligned}
\nabla_\alpha F(x) &= \sum_{i=1}^{N} f_i \nabla_\alpha M_{d_i}(x) \\
&= \sum_{i=1}^{N} f_i \left(-\sum_{j=0}^{\ell} d_{i,j} \alpha^{d_i - p^j} x^{p^j} \right) \\
&= \sum_{j=0}^{\ell} \left(\sum_{i=1}^{N} -f_i d_{i,j} \alpha^{d_i - p^j} \right) x^{p^j} \\
&= \sum_{j=0}^{\ell} g_j^{(F)}(\alpha) x^{p^j}.
\end{aligned}$$

□

4.1 Cryptographic properties of GDO functions

In this section, we discuss the differential uniformity and the Walsh coefficient of GDO functions. Although neither of these notions is preserved by GEA-equivalence, they are both relevant cryptographic criteria in the context of differential [4] and linear [20] cryptanalysis of block ciphers, respectively. Moreover, the homogeneity of GDO functions leads to a simplification of this cryptographic analysis.

First, let us state the following proposition.

Proposition 7. *Let $F : \mathbb{F}_{p^n} \to \mathbb{F}_{p^n}$ be a GDO function. Then, $F(ix) = iF(x)$ for all $i \in \mathbb{F}_p$.*

Definition 11 (Differential uniformity)**.** The differential uniformity of a function $F : \mathbb{F}_{p^n} \to \mathbb{F}_{p^n}$ is the value:

$$\delta_F = \max_{\alpha \neq 0, \beta \in \mathbb{F}_{p^n}} \# \left\{ x \in \mathbb{F}_{p^n} \mid \Delta_\alpha F(x) = \beta \right\}.$$

Theorem 8. *Let $F : \mathbb{F}_{p^n} \to \mathbb{F}_{p^n}$ be a GDO function. Then, $\delta_F \geq p$.*

Proof. Let $\alpha \in \mathbb{F}_{p^n}^*$. For $j \in \mathbb{F}_p$, and from Proposition 7, we have:

$$\Delta_\alpha F(j\alpha) = F(j\alpha + \alpha) - F(j\alpha) = (j + 1 - j)F(\alpha) = F(\alpha).$$

Thus, the equation $\Delta_\alpha F(x) = F(\alpha)$ has at least p solutions, for any $\alpha \in \mathbb{F}_{p^n}$. □

Remark 3. Functions that are GEA-equivalent to GDO functions can have a differential uniformity strictly less than p (by choosing the function B in Definition 9 with an algebraic degree less than or equal to $p-1$). For example, if we take $p = 5$, $n = 2$ and $F(x) = x^{p+p-1} + x^{p-3} \in \mathbb{F}_{p^n}[x]$, then:

- F is a GAPN function of algebraic degree p.
- We can verify (e.g. using Sagemath) that $\delta_F = 4 = p - 1$.

Definition 12. Let p be a prime number, $n \geq 0$ be an integer, and ω_p be a p-root of unity of $\mathbb{C}$. The Walsh-coefficients of a function $F : \mathbb{F}_{p^n} \to \mathbb{F}_{p^n}$ are

$$W_F(a,b) = \sum_{x \in \mathbb{F}_{p^n}} \omega_p^{Tr(aF(x)-bx)}, \quad \forall (a,b) \in \mathbb{F}_{p^n}^* \times \mathbb{F}_{p^n},$$

where $Tr(x) = \sum_{i=0}^{n-1} x^{p^i}$ is the absolute trace from $\mathbb{F}_{p^n}$ to $\mathbb{F}_p$.

Theorem 9. *Let $F : \mathbb{F}_{p^n} \to \mathbb{F}_{p^n}$ be a GDO function. Then,*

$$W_F(a,b) := A_F^{(a,b)}(0) - A_F^{(a,b)}(1) \in \mathbb{Z}, \quad \forall\, (a,b) \in \mathbb{F}_{p^n}^* \times \mathbb{F}_{p^n},$$

where $A_F^{(a,b)}(c) = \#\{x \in \mathbb{F}_{p^n} \mid Tr(aF(x) - bx) = c\}$ and $c \in \mathbb{F}_p$.

Proof. First, we have

$$\begin{aligned} W_F(a,b) &= \sum_{x \in \mathbb{F}_{p^n}} \omega_p^{Tr(aF(x)-bx)} \\ &= \sum_{c \in \mathbb{F}_p} \omega_p^c A_F^{(a,b)}(c). \end{aligned}$$

Secondly, since F is a homogeneous function of algebraic degree p, from Proposition 7 we have $Tr(aF(x) - bx) = c \Leftrightarrow Tr(aF(ix) - bix) = ic$ for all $(i,c) \in \mathbb{F}_p^* \times \mathbb{F}_p$. In other words, $A_F^{(a,b)}(c) = A_F^{(a,b)}(1)$ for all $c \in \mathbb{F}_p^*$. Finally, since $0 = \sum_{c \in \mathbb{F}_p} \omega_p^c = 1 + \sum_{c \in \mathbb{F}_p^*} \omega_p^c$, we obtain

$$\begin{aligned} W_F(a,b) &= \sum_{x \in \mathbb{F}_{p^n}} \omega_p^{Tr(aF(x)-bx)} = A_F^{(a,b)}(0) + \sum_{c \in \mathbb{F}_p^*} A_F^{(a,b)}(c)\omega_p^c \\ &= A_F^{(a,b)}(0) + A_F^{(a,b)}(1) \sum_{c \in \mathbb{F}_p^*} \omega_p^c = A_F^{(a,b)}(0) - A_F^{(a,b)}(1). \end{aligned}$$

□

Remark 4. A similar result can be obtained for homogeneous functions of algebraic degree a power of p, the field's characteristic.

Corollary 1. *Let $F : \mathbb{F}_{p^n} \to \mathbb{F}_{p^n}$ be a GDO function. Then,*

$$W_F(a,b) = p^n - p \times A_F^{(a,b)}(1), \quad \forall\, (a,b) \in \mathbb{F}_{p^n} \times \mathbb{F}_{p^n}^*.$$

Proof. We know that $p^n = \sum_{c \in \mathbb{F}_p} A_F^{(a,b)}(c) = A_F^{(a,b)}(0) + (p-1) \times A_F^{(a,b)}(1)$. The result follows from Theorem 9. □

Remark 5. For a GDO function $F : \mathbb{F}_{p^n} \to \mathbb{F}_{p^n}$, we have the following property

$$A_F^{(a,b)}(c) = A_F^{(ia,ib)}(c), \quad \forall\, (a,b) \in \mathbb{F}_{p^n}^* \times \mathbb{F}_{p^n}, \forall\, (i,c) \in \mathbb{F}_p^* \times \mathbb{F}_p.$$

4.2 Preliminary results

Before we proceed further into the characterization of GAPN classes of GDO polynomials, we need to introduce a couple of lemmas that require the concept of modular order of an integer.

Definition 13 (Modular order of an integer)**.** We call the order of an integer i modulo n, the smallest strictly positive integer, denoted $\mathbb{O}_n(i)$, satisfying

$$i^{\mathbb{O}_n(i)} \equiv 1 \pmod{n}.$$

Lemma 10. *Let $n \geq 2$ be an integer, $d \neq 1$ be a divisor of n, and p be a prime number coprime to n. Then, $\mathbb{O}_d(p) | \mathbb{O}_n(p)$. In particular $\mathbb{O}_d(p) \leq \mathbb{O}_n(p)$.*

Proof. If $n = kd$ for some integer $k \geq 1$, then:

$$\begin{aligned} p^{\mathbb{O}_n(p)} \equiv 1 \pmod{n} &\implies \exists\, m \in \mathbb{N}^*,\ p^{\mathbb{O}_n(p)} = 1 + nm \\ &\implies p^{\mathbb{O}_n(p)} = 1 + kdm \\ &\implies p^{\mathbb{O}_n(p)} \equiv 1 \pmod{d}. \end{aligned}$$

Using Lagrange's theorem, we have $\mathbb{O}_d(p) | \mathbb{O}_n(p)$ as $\{p^i \pmod{d} \mid i \in \mathbb{N}\}$ is a cyclic group for multiplication of order $\mathbb{O}_d(p)$. Particularly, we have $\mathbb{O}_d(p) \leq \mathbb{O}_n(p)$. □

Lemma 11. *Let $n = q_1^{i_1} q_2^{i_2} \cdots q_s^{i_s}$ be the prime factorization of an integer n, for some prime numbers $q_1, \ldots, q_s$ and some positive exponents $i_1, \ldots, i_s \geq 1$. Let p be a prime number that does not divide n. Then,*

$$\min_{i \in \{1,\ldots,s\}} \mathbb{O}_{q_i}(p) = \min_{d > 1, d|n} \mathbb{O}_d(p).$$

Proof. If $d|n$, then $\exists\, i \in \{1, \ldots, s\}$ such that $q_i | d$. By Lemma 10, we obtain $\mathbb{O}_{q_i}(p) \leq \mathbb{O}_d(p)$. In particular, $\min\limits_{i \in \{1,\ldots,s\}} \mathbb{O}_{q_i}(p) \leq \mathbb{O}_d(p)$ for every divisor $d > 1$ of n, i.e.,

$$\min_{i \in \{1,\ldots,s\}} \mathbb{O}_{q_i}(p) \leq \min_{d > 1, d|n} \mathbb{O}_d(p).$$

As $\{\mathbb{O}_{q_1}(p), \ldots, \mathbb{O}_{q_s}(p)\} \subset \{\mathbb{O}_d(p) \mid d > 1 \text{ divides } n\}$, we have

$$\min_{i \in \{1,\ldots,s\}} \mathbb{O}_{q_i}(p) = \min_{d > 1, d|n} \mathbb{O}_d(p).$$

□

Lemma 12. *Let q and p be two different prime numbers. We have, for all integers $n, m \geq 1$,*

$$\mathbb{O}_q(p^n) \leq \mathbb{O}_q(p) \leq \mathbb{O}_{q^m}(p).$$

If $u_{m,n}(q,p) := \mathbb{O}_{q^m}(p^n)$, then $(u_{m,n}(q,p))_m$ is an increasing sequence for all integers $n \geq 1$, and, $(u_{m,n}(q,p))_n$ is a decreasing sequence for all integers $m \geq 1$.

Proof.

- In fact, $\mathbb{O}_q(p^n)$ divides $\mathbb{O}_q(p)$:

$$(p^n)^{\mathbb{O}_q(p)} = (p^{\mathbb{O}_q(p)})^n \equiv 1^n \pmod{q} = 1 \pmod{q}.$$

Applying the same argument as in the proof of Lemma 10, we get $\mathbb{O}_q(p^n) \leq \mathbb{O}_q(p)$.

- The inequality $\mathbb{O}_q(p) \leq \mathbb{O}_{q^m}(p)$ follows directly from Lemma 10.

$\square$

Since our study also makes use of cyclotomic polynomials, we recall the following definition from [19].

Definition 14 (**d-th cyclotomic polynomial**)**.** Let d be a positive integer not divisible by a prime p and ζ a primitive d-th root of unity over $\mathbb{F}_{p^n}$. Then the polynomial

$$Q_d(x) := \prod_{\substack{s=1,\ldots,d \\ gcd(s,d)=1}} (x - \zeta^s)$$

is called the d-th cyclotomic polynomial over $\mathbb{F}_{p^n}$.

Later, we will use the following result, inspired directly by [19, Theorem 2.47].

Lemma 13. *Let $Q_d(x) \in \mathbb{F}_p[x]$ be the d-th cyclotomic polynomial such that $d \not\equiv 0 \pmod p$. Then $Q_d(x)$ factors into $\phi(d)/\mathbb{O}_d(p)$ distinct monic irreducible polynomials in $\mathbb{F}_p[x]$ of the same polynomial degree $\mathbb{O}_d(p)$, where ϕ is the Euler totient function.*

We also need the following necessary and sufficient criteria for monomial functions of the GDO type to be GAPN functions, due to Kuroda [16]. However, we instead use the equivalent formulation due to Ozbudak and Salagean [24], which is more convenient for our purposes.

Lemma 14 ([24] Theorem 1)**.** *Let $F(x) = x^e \in \mathbb{F}_{p^n}[x]$ be a GDO polynomial, where $e = \sum_{i=0}^{\ell} k_i.p^i$, $k_0 \neq 0$ and $k_\ell \neq 0$. Then, F is a GAPN function over $\mathbb{F}_{p^n}$ if, and only if,*

$$gcd\left(\sum_{i=0}^{\ell} k_i.z^i, z^n - 1\right) = z - 1 \quad \text{in} \quad \mathbb{F}_p[z].$$

5 New classes of GAPN functions

5.1 GDO monomials

In [16], Kuroda showed that every GDO monomial over an extension field $\mathbb{F}_{p^n}$ is a GAPN function over some extension of $\mathbb{F}_{p^n}$. He also showed that every GDO monomial is a p-exceptional GAPN function, i.e., it is a GAPN function over an infinite number of extension fields $\mathbb{F}_{p^n}$. Table 1 recapitulates all known GDO monomials, which are GAPN functions, including our results. In what follows, we give a way to find, for every GDO monomial function, an infinite number of extensions, with an explicit formula for the extension degree, where this function is GAPN.

This first result gives sufficient conditions for a GDO monomial to be GAPN.

Theorem 15. *Let $e = \sum_{i=0}^{\ell} k_i p^i$ be an exponent such that $F(x) = x^e$ is of the GDO type over $\mathbb{F}_{p^n}$, and, let $n = q_1^{i_1} \cdots q_s^{i_s}$ be the prime factorization of n. We have two cases:*

1. *$n \not\equiv 0 \pmod p$: If $\ell \leq \min_{1 \leq i \leq s} \mathbb{O}_{q_i}(p)$, then F is a GAPN function.*

2. *$n \equiv 0 \pmod p$: If $\ell \leq \min\limits_{1\leq i\leq s,\ q_i\neq p} \mathbb{O}_{q_i}(p)$ or $n = p^\alpha$, and if the multiplicity of 1 as a root in $\sum_{i=0}^{\ell} k_i x^i \in \mathbb{F}_p[x]$ is equal to 1, then F is a GAPN function.*

Proof. Let $g(x) = \sum_{i=0}^{\ell} k_i x^i \in \mathbb{F}_p[x]$, and let $R(x) := gcd\,(g(x), x^n - 1) \in \mathbb{F}_p[x]$. From Lemma 14, F is a GAPN function if, and only if, $R(x) = x - 1$.

1. When $n \not\equiv 0 \pmod p$, we have (see [19, Theorem 2.45])

$$x^n - 1 = (x-1) \prod_{d>1,\ d|n} Q_d(x),$$

where $Q_d(x) \in \mathbb{F}_p[x]$ is the d-th cyclotomic polynomial (see Definition 14).

Since $g(1) = \sum_{i=0}^{\ell} k_i = p \equiv 0 \pmod p$, we can factorize $g(x)$ by $(x-1)^m$ where $m \geq 1$ is the multiplicity of 1 as a root of $g(x)$. In other words, there exists $V(x) \in \mathbb{F}_p[x]$, $V(1) \neq 0$, such that

$$g(x) = (x-1)^m \times V(x).$$

Hence,

$$R(x) = (x-1) \times gcd\left(V(x), \prod_{d>1,\ d|n} Q_d(x)\right).$$

A sufficient condition for $R(x)$ to be equal to $x - 1$ is to force every irreducible factor of $\prod_{d>1,\ d|n} Q_d(x)$ in $\mathbb{F}_p[x]$ to have a degree strictly greater than the degree of $V(x)$. We ensure it using Lemma 13, i.e., we must have

$$\mathbb{O}_d(p) > \deg\left(V(x)\right) = \ell - m$$

for every $d > 1$ dividing n. In particular, if $\mathbb{O}_d(p) \geq \ell$, then we have $\mathbb{O}_d(p) > \ell - m$ because $m > 0$.
Therefor, if

$$\ell \leq \min\left(\{\mathbb{O}_d(p) \mid d > 1 \text{ divide } n\}\right),$$

then F is a GAPN function over $\mathbb{F}_{p^n}$.
Using Lemma 11, we conclude $\min(\{\mathbb{O}_d(p) \mid d > 1 \text{ divide } n\}) = \min\limits_{1\leq i\leq s} \mathbb{O}_{q_i}(p)$.

2. In this case, we can write $n = v.p^u$ where $gcd(p, v) = 1$ and $u \in \mathbb{N}^*$. We have

$$x^n - 1 = (x^v - 1)^{p^u},$$

i.e.,

$$x^n - 1 = (x-1)^{p^u} \prod_{d>1, d|v} Q_d(x)^{p^u}.$$

When the multiplicity of 1 in $g(x)$ is equal to 1 and when the degree of $Q_d(X)$ for $d > 1$ and $d|n$ is strictly greater than $\deg(V) \leq \ell - 1$, we get $R(x) = x - 1$.

□

Remark 6. The result of Theorem 15 above remains true when $p = 2$.

Proposition 16. *Let p be a prime number and n be an integer with prime factorization*

$$n = \prod_{j=1}^{s} q_j^{i_j} \not\equiv 0 \pmod{p},$$

where $q_1, \ldots, q_s$ are distinct primes and $i_1, \ldots, i_s \geq 1$, and assume that $p \nmid n$. For every integer $\ell \leq n-1$, if $\min_{1 \leq i \leq s} \mathbb{O}_{q_i}(p) \geq \ell$, then $gcd(n, \ell) = 1$.

Proof. We always have:

$$gcd(x^\ell - 1, x^n - 1) = x^{gcd(\ell,n)} - 1 \in \mathbb{F}_p[x].$$

And, with the conditions in Theorem 15, we have $gcd(x^\ell - 1, x^n - 1) = x - 1 \in \mathbb{F}_p[x]$. □

Remark 7. Proposition 16 shows that when $p = 2$, some of the Gold functions, i.e., quadratic APN monomial functions [23], are obtained by our method in Theorem 15.

We can illustrate the first point of Theorem 15 by giving some examples. We first fix an odd prime number p and an integer $\ell \geq 2$. We then find some integers $k_0, \ldots, k_\ell$ such that $0 \leq k_i \leq p-1$ for every $0 \leq i \leq \ell$ and $\sum_{i=0}^{\ell} k_i = p$. Let $e = \sum_{i=0}^{\ell} k_i p^i$. From Theorem 15, we can find precisely some of the extensions where $F(x) = x^e$ is a GAPN function.

Example 2. Let $p = 5$ and $\ell = 2$. The functions $F(x) = x^e$, where

$$e \in \left\{p^2 + p + 3, p^2 + 2p + 2, 3p^2 + p + 1, 2p^2 + p + 2\right\},$$

are all GAPN over $\mathbb{F}_{5^n}$ when $n = q_1^{i_1} \cdots q_s^{i_s}$, where $q_1, \ldots, q_s$ are odd primes different from 5 and are such that $\min_{1 \leq i \leq s} \mathbb{O}_{q_i}(5) \geq 2$. As a simple case, they are GAPN functions over $\mathbb{F}_{5^3}$ because $\mathbb{O}_3(5) = 2 = \ell$.

We can use the second point of Theorem 15 to construct GAPN functions as well. We first fix a prime number p and an integer $\ell \geq 2$. Then, we find $k_1, \ldots, k_\ell$ such that $0 \leq k_i \leq p-1$ for every $0 \leq i \leq \ell$ and $\sum_{i=0}^{\ell} k_i = p$. We are looking for extensions $\mathbb{F}_{p^n}$ where the GDO monomial $F(x) = x^{\sum_{i=0}^{\ell} k_i p^i}$ is GAPN. Let us denote

$$V(x) := \frac{\sum_{i=0}^{\ell} k_i . x^i}{x - 1} \in \mathbb{F}_p[x],$$

and, suppose that it satisfies $V(1) \neq 0$, i.e., 1 has a multiplicity exactly one as a root of $\sum_{i=0}^{\ell} k_i x^i \in \mathbb{F}_p[x]$. Then, from the point 2 of Theorem 15, we can find extensions of $\mathbb{F}_p$ where the GDO function $F(x) = x^{\sum_{i=0}^{\ell} k_i . p^i}$ is GAPN. For example:

Example 3. Let $p = 7$ and $\ell = 4$. Using the same notations as in Theorem 15:

1. Let $g(x) = x^4 + 2x^3 + 3x^2 + 1 \in \mathbb{F}_7[x]$. We have:
   - $g(1) = 1 + 2 + 3 + 1 = 7 = 0$,

- $V(x) = \frac{g(x)}{x-1} = x^3 + 3x^2 + 6x + 6$,
- $V(1) = 1 + 3 + 6 + 6 = 16 = 2 \neq 0$.

2. Let $g(x) = 2x^4 + 3x^3 + x^2 + 1 \in \mathbb{F}_7[x]$. We have:

   - $g(1) = 2 + 3 + 1 + 1 = 7 = 0$,
   - $V(x) = \frac{g(x)}{x-1} = 2x^3 + 5x^2 + 6x + 6$,
   - $V(1) = 2 + 5 + 6 + 6 = 19 = 5 \neq 0$.

We conclude that the functions $F(x) = x^e$, where

$$e \in \left\{p^4 + 2p^3 + 3p^2 + 1(= 3235), 2p^4 + 3p^3 + p^2 + 1(= 5881)\right\},$$

are GAPN over $\mathbb{F}_{7^{7^\alpha \times N}}$ with $N = q_1^{i_1} \cdots q_s^{i_s}$, where the primes $q_1, \ldots, q_s$ are different from 7, $(i_1, \ldots, i_s) \in \mathbb{N}^s$, and such that, $i_1 = \cdots = i_s = 0$ or $\min\limits_{1 \leq i \leq s} \mathbb{O}_{q_i}(7) \geq 4$.
As a simple case, they are GAPN over $\mathbb{F}_{7^{7\times 5}} : \mathbb{O}_5(7) = 4 = \ell$. From Theorem 15, we can confirm that these functions are GAPN over all the extensions $\mathbb{F}_{7^{7^\alpha \times 5^\beta}}$ for every $(\alpha, \beta) \in \mathbb{N}^* \times \mathbb{N}$.

We now provide some examples in the case $p = 2$. The first one illustrates the first point of Theorem 15 while the second illustrates the second point of the same theorem.

Example 4. Let $\ell = 3$.

- Let $F(x) = x^{2^3+1}$. Then, using the first point of Theorem 15, we only have to find odd primes $q_1, \ldots, q_s$ such that $\min\limits_{1 \leq i \leq s} \mathbb{O}_{q_i}(2) \geq 3$, and we will be sure that F is APN over any extension $\mathbb{F}_{2^n}$ where $n = q_1^{i_1} \cdots q_s^{i_s}$ and $(i_1, \ldots, i_s) \neq (0, \ldots, 0)$. For example, if we take $n = 7$, then F is APN over $\mathbb{F}_{2^7}$ (more generally, over $\mathbb{F}_{2^{7^\alpha}}$, $\alpha \in \mathbb{N}^*$): In fact, $\mathbb{O}_7(2) = 6 \geq 3$.

- Let $F(x) = x^{2^3+1}$. Then:

  - $g(x) = x^3 + 1$,
  - $g(1) = 2 = 0$,
  - $V(x) = x^2 + x + 1$,
  - $V(1) = 1 + 1 + 1 = 3 = 1 \neq 0$,

  We conclude that F is APN over $\mathbb{F}_{2^{2^\alpha . n}}$ where $\alpha \in \mathbb{N}^*$, $n \in \{1, q_1^{i_1} \times \cdots \times q_s^{i_s}\}$ and such that $\min\limits_{1 \leq i \leq s} \mathbb{O}_{q_i}(2) \geq 3$.

To give a sufficient condition for a GDO monomial to be a GAPN function over $\mathbb{F}_{p^n}$ when n is even and p is odd, one sees that the exponent must satisfy some additional criteria. First, we state the following lemma.

Lemma 17. *Let p be an odd prime number, and consider the polynomial $g(x) = \sum_{i=0}^{\ell} k_i x^i \in \mathbb{N}[x]$ where $0 \leq k_i \leq p-1$ for all $0 \leq i \leq \ell$ and $\sum_{i=0}^{\ell} k_i = p$. Then, $g(-1) = 0 \pmod p$ if, and only if, either $k_{2j} = 0$ for all $j \geq 0$ or $k_{2j+1} = 0$ for all $j \geq 0$.*

Proof. First, suppose that $k_{2j} = 0$ for all $j \geq 0$. Then

$$g(-1) = \sum_{i=0}^{\ell} k_i(-1)^i = \sum_{i=0,\ i\ odd}^{\ell} k_i(-1)^i$$
$$= -\sum_{i=0,\ i\ odd}^{\ell} k_i = -\sum_{i=0}^{\ell} k_i = -p = 0 \pmod p.$$

Now, if $k_{2j+1} = 0$ for all $j \geq 0$, then

$$g(-1) = \sum_{i=0}^{\ell} k_i(-1)^i = \sum_{i=0,\ i\ even}^{\ell} k_i(-1)^i = \sum_{i=0,\ i\ even}^{\ell} k_i$$
$$= \sum_{i=0}^{\ell} k_i = p = 0 \pmod p.$$

On the other hand, if $g(-1) = 0 \pmod p$, then

$$0 \pmod p = g(-1) = \sum_{i=0}^{\ell} k_i(-1)^i = \sum_{i=0,\ i\ even}^{\ell} k_i - \sum_{i=0,\ i\ odd}^{\ell} k_i.$$

Moreover, we have

$$p = \sum_{i=0}^{\ell} k_i = \sum_{i=0,\ i\ even}^{\ell} k_i + \sum_{i=0,\ i\ odd}^{\ell} k_i = 2 \times \sum_{i=0,\ i\ odd}^{\ell} k_i \pmod p.$$

The equality above means that p divides $2 \times \sum_{i=0,\ i\ odd}^{\ell} k_i$, but since p is an odd prime number, p necessarily divides $\sum_{i=0,\ i\ odd}^{\ell} k_i$. We get two cases: either $\sum_{i=0,\ i\ odd}^{\ell} k_i = 0$, i.e. $k_i = 0$ for all odd $0 \leq i \leq \ell$, or $\sum_{i=0,\ i\ odd}^{\ell} k_i = p$, hence $\sum_{i=0,\ i\ even}^{\ell} k_i = 0$. □

The following theorem gives other sufficient conditions for a GDO monomial function to be GAPN, that are not covered by Theorem 15, notably when the degree extension of the field is even.

Theorem 18. *Let p be an odd prime, and, let $n = 2N \not\equiv 0 \pmod p$ where $N \geq 3$ is an odd integer. Let $e = \sum_{i=0}^{\ell} k_i p^i$ be an exponent such that $F(x) = x^e$ is of the GDO type, and where there exist $0 \leq i, j \leq \ell$ such that $k_{2i} \neq 0$ and $k_{2j+1} \neq 0$. If*

$$N = q_2^{i_2} \cdots q_m^{i_m} \text{ and } \ell \leq \min_{2 \leq i \leq m} \mathbb{O}_{q_i}(p),$$

then $F(x) = x^e$ is a GAPN function over $\mathbb{F}_{p^n}$.

Proof. Recall that $n = 2N$. Suppose that $\ell \leq \min_{2 \leq i \leq m} \mathbb{O}_{q_i}(p)$.

Let $g(x) = \sum_{i=0}^{\ell} k_i x^i \in \mathbb{F}_p[x]$. We have that

$$x^n - 1 = x^{2N} - 1 = (x-1)(x+1) \prod_{d|N,\ d>1} Q_{2d}(x).$$

Since $g(1) = \sum_{i=0}^{\ell} k_i = p \equiv 0 \pmod p$, we can factorize $g(x)$ by $(x-1)^m$ where $m \geq 1$ is the multiplicity of 1 as a root of $g(x)$. In other words, there exists $V(x) \in \mathbb{F}_p[x]$, with $V(1) \neq 0$, such that

$$g(x) = (x-1)^m \times V(x).$$

Hence, as $V(-1) \neq 0$ because of the choice of e (Lemma 17), we have:

$$gcd(g(x), x^{2N} - 1) = (x-1) \times gcd\left(V(x), \prod_{d|N,\ d>1} Q_{2d}(x)\right).$$

From Lemma 10, we have that $\mathbb{O}_{2d}(p) \geq \mathbb{O}_d(p) \geq \ell > \deg(V(x))$ for every divisor $d \neq 1$ of N, i.e.,

$$\begin{aligned} gcd(g(x), x^{2N} - 1) &= (x-1) \times gcd\left(V(x), \prod_{d|N,\ d>1} Q_{2d}(x)\right) \\ &= (x-1). \end{aligned}$$

Applying Lemma 14 concludes the proof. □

Remark 8. This approach cannot be generalized to $n = 2^a N$ with $a > 1$. The only result that we can get from this approach is that $F(x) = x^{ip+p-i}$ is always GAPN for every $1 \leq i \leq p-1$, which is already known (special case of [27, Theorem 2]). This is because $\mathbb{O}_2(p) = 1$ for any odd prime number p.

Our results in Theorem 15 and Theorem 18 are independent of the multiplicity m of 1 as a root in $g(x) = \sum_{i=0}^{\ell} k_i x^i \in \mathbb{F}_p[x]$, except for the second point of Theorem 15. But as we showed in the proofs of these two theorems, we can obtain additional extensions in which a GDO monomial is GAPN by requiring $\min_{2 \leq i \leq \ell} \mathbb{O}_{q_i}(p) \geq \ell - m + 1$, which depends on m, the multiplicity of 1 as root of $g(x)$. For example, we have the following result.

Corollary 2. *Let $g(x) = \sum_{i=0}^{\ell} k_i.x^i \in \mathbb{Z}[x]$ be such that $\sum_{i=0}^{\ell} k_i = p$ (where $0 \leq k_i \leq p-1$). If the multiplicity of 1 as a root in $g \pmod p$ is ℓ, then $F(x) = x^{g(p)}$ is a GAPN function over $\mathbb{F}_{p^n}$ for $n > \ell$ if, and only if, $n \not\equiv 0 \pmod p$.*

Proof. Let $F(x) = x^{g(p)}$ and $R(z) = gcd\,(g(z), z^n - 1) \in \mathbb{F}_p[z]$ where $n > \ell$. F is a GAPN function over $\mathbb{F}_{p^n}$ if, and only if,

$$R(z) = z - 1.$$

As $g(z) = k_\ell (z-1)^\ell$, we have $R(z) = gcd\left((z-1)^\ell, z^n - 1\right)$. And, $R(z) = z - 1 \in \mathbb{F}_p[z]$ if, and only if, the multiplicity of 1 as a root in $z^n - 1$ is exactly 1, i.e., if, and only if, $n \not\equiv 0 \pmod p$. □

Example 5. The functions $F(x) = x^{ip^2+(p-2i)p+i}$, where $i \in \{1, \ldots, \frac{p-1}{2}\}$, are GAPN over $\mathbb{F}_{p^n}$ for every $n \geq 3$ such that $n \not\equiv 0 \pmod p$.

Example 6. The function $F(x) = x^{1+p+\cdots+p^{p-1}}$ is a GAPN function over $\mathbb{F}_{p^n}$ for every $n \geq p$ such that $n \not\equiv 0 \pmod p$.
In fact, $(x-1)^p = x^p - 1 = (x-1)(1 + x + \cdots + x^{p-1})$, i.e., $1 + x + \cdots + x^{p-1} = (x-1)^{p-1}$, i.e., the multiplicity of 1 as a root of $1 + x + \cdots + x^{p-1}$ is exactly $p-1$.

Remark 9. An important thing about Theorem 15 and Theorem 18 is that we only care about the primes $q_1, \ldots, q_s$ in the prime factorization of n. If these prime numbers satisfy the conditions, then the same function will be GAPN over any extension $\mathbb{F}_{p^m}$ where $m = q_1^{i_1} \cdots q_s^{i_s}$ for every $(i_1, \ldots, i_s) \in \mathbb{N}^s \backslash \{(0, \ldots, 0)\}$.

In the next theorem, we present a specific case of GDO monomials with a low polynomial degree and their sufficient and necessary conditions to be GAPN functions.

Theorem 19. *Let $F(x) = x^e$ a function over $\mathbb{F}_{p^n}$ where $e = a_2p^2 + a_1p + a_0$ and $a_2 + a_1 + a_0 = p$. Let $n = p^\alpha \times N$ where $\alpha \geq 0$ and $gcd(N, p) = 1$.*
Then, F is GAPN over $\mathbb{F}_{p^n}$ if, and only if, either

- $a_0^N \neq a_2^N$, *or,*
- $a_0 = a_2$ *and* $\alpha = 0$.

Proof. The p-polynomial $\nabla_1 F(x)$ has conventional p-associate polynomial (see Definition 6):

$$\begin{aligned} g(x) &= -\left(a_2x^2 - (a_0 + a_2)x + a_0\right) \\ &= -a_2(x-1)(x - \frac{a_0}{a_2}). \end{aligned}$$

From [26, Theorem 1], we know that F is a GAPN function over $\mathbb{F}_{p^n}$ if, and only if,

$$gcd\left(x^n - 1, g(x)\right) = x - 1$$

in $\mathbb{F}_p[x]$.

- If $n \not\equiv 0 \pmod p$: $x^n - 1 = (x-1) \times \prod_{d|n \ d>1} Q_d(x)$. So, the condition is satisfied if, and only if, $\frac{a_0}{a_2}$ is not an n-th root of unity different from 1.
- If $n \equiv 0 \pmod p$, then $x^n - 1 = (x-1)^{p^u} . \prod_{d|v \ d>1} Q_d(x)^{p^u}$ where $n = vp^u$ and $gcd(v, p) = 1$. In this case, the condition in [27, Theorem 1] is satisfied if, and only if, 1 is a simple root of g and $\frac{a_0}{a_2}$ is not a v-root of unity.

□

When their Hamming degree differ, we can be sure that most of our GAPN monomial functions are GEA-inequivalent to existing GAPN monomial functions. More precisely [28, Proposition 3], two monomial functions $F_1, F_2 : \mathbb{F}_{p^n} \to \mathbb{F}_{p^n}$ with the same algebraic degree are GEA-equivalent if, and only if, there exist $0 \leq i, j \leq n-1$ and $\alpha, \beta \in \mathbb{F}_{p^n}^*$ such that $F_1(X) = \alpha F_2(\beta X^{p^i})^{p^j}$. In other words, if two monomial functions are GEA-equivalent, then they must have the same Hamming degree.

5.2 GDO multinomials

We are now interested in the GAPN-ness of non-monomial GDO functions. As already stated, to know if a function represented by a GDO polynomial is a GAPN function, one might study the kernel of its generalized derivative functions, which are linearized functions.

Hence, we first recall a result of McGuire and Sheekey [21] on the number of roots of linearized polynomials over extensions of $\mathbb{F}_p$. They give a full description of these roots. From this result, we draw classes of non-monomial GAPN functions with algebraic degree p.

Theorem 20 ([21] Theorem 5)**.** *Let $L(x) = \sum_{i=0}^{\ell} a_i x^{p^i} \in \mathbb{F}_{p^n}[x]$ be a linearized polynomial where $a_\ell \neq 0$. Let*

$$C^{(p^i)}(L) := \begin{bmatrix} 0 & 0 & \dots & 0 & -\left(\frac{a_0}{a_\ell}\right)^{p^i} \\ 1 & 0 & \dots & 0 & -\left(\frac{a_1}{a_\ell}\right)^{p^i} \\ \vdots & \vdots & \ddots & 0 & \vdots \\ 0 & 0 & \dots & 1 & -\left(\frac{a_{\ell-1}}{a_\ell}\right)^{p^i} \end{bmatrix}, \quad i \in \{0, 1, \dots, n-1\}$$

and

$$A(L) := \prod_{i=0}^{n-1} C^{(p^i)}(L).$$

Then, the number of roots of L in $\mathbb{F}_{p^n}$ is exactly p^d where $d = \dim_{\mathbb{F}_p}(ker(A(L) - I_\ell))$, and I_ℓ is the identity matrix of size ℓ.

Using the results and notation from Lemma 6 about the coefficients of the generalized derivative functions of a GDO polynomial, and Theorem 20 above about the roots of these linearized derivative functions, we have the following characterization.

Theorem 21. *Let p be an odd prime number and $F(x) = \sum_{i=1}^{N} f_i.x^{d_i}$ be a GDO function over $\mathbb{F}_{p^n}$ where $d_i = \sum_{j=0}^{\ell} d_{i,j} p^j$. Suppose that $g_1^{(F)}, \dots, g_\ell^{(F)}$, defined in Lemma 6 as the polynomials representing the coefficients of the generalized derivatives of F, have no nonzero root in common. Then, F is GAPN over $\mathbb{F}_{p^n}$ if, and only if, the following condition holds:*

- *$\forall\, \alpha \in \mathbb{F}_{p^n}^*$, let $t_\alpha \in \{0, \dots, \ell\}$ be the greatest integer such that $g_{t_\alpha}^{(F)}(\alpha) \neq 0$. The condition is:*

  $$\dim_{\mathbb{F}_p}\left(\ker(A(\nabla_\alpha F) - I_{t_\alpha})\right) = 1 \quad \forall\, \alpha \in \mathbb{F}_{p^n}^*$$

  where

  $$A(\nabla_\alpha F) := \prod_{i=0}^{n-1} C^{(p^i)}(\nabla_\alpha F)$$

  and

  $$C^{(p^i)}(\nabla_\alpha F) = \begin{bmatrix} 0 & 0 & \dots & 0 & -\left(\frac{g_0^{(F)}(\alpha)}{g_{t_\alpha}^{(F)}(\alpha)}\right)^{p^i} \\ 1 & 0 & \dots & 0 & -\left(\frac{g_1^{(F)}(\alpha)}{g_{t_\alpha}^{(F)}(\alpha)}\right)^{p^i} \\ & & \ddots & & \vdots \\ 0 & 0 & \dots & 1 & -\left(\frac{g_{t_\alpha-1}^{(F)}(\alpha)}{g_{t_\alpha}^{(F)}(\alpha)}\right)^{p^i} \end{bmatrix} \in \mathbb{F}_{p^n}^{t_\alpha \times t_\alpha}.$$

Proof. Let $\alpha \in \mathbb{F}_{p^n}^*$. From Lemma 6 we have

$$\nabla_\alpha F(x) = \sum_{j=0}^{\ell} g_j^{(F)}(\alpha) x^{p^j}.$$

If $g_0^{(F)},\ g_1^{(F)}, \dots,\ g_\ell^{(F)}$ have a common root $\alpha \neq 0$, then we have $\nabla_\alpha F(x) = 0$ for every $x \in \mathbb{F}_{p^n}$, i.e., F is not GAPN.
Otherwise, if $g_0^{(F)},\ g_1^{(F)}, \dots,\ g_\ell^{(F)}$ have no nonzero common root, then for every $\alpha \in \mathbb{F}_{p^n}^*$, there exists $t_\alpha \in \{0, 1, \dots, \ell\}$ such that t_α is the greatest possible value satisfying $g_{t_\alpha}^{(F)}(\alpha) \neq 0$ (else, $g_i^{(F)}(\alpha) = 0$ for all $0 \leq i \leq \ell$, which contradict the no nonzero common root assumption).
Therefore, we can rewrite

$$\nabla_\alpha F(x) = \sum_{j=0}^{t_\alpha} g_j^{(F)}(\alpha) x^{p^j}.$$

We can then apply Theorem 20, for any $\alpha \in \mathbb{F}_{p^n}^*$, to say that $\nabla_\alpha F(x)$ has p roots in $\mathbb{F}_{p^n}$ if, and only if,

$$\dim_{\mathbb{F}_p}(\ker(A(\nabla_\alpha F) - I_{t_\alpha})) = 1.$$

□

Remark that Theorem 21 applies to any GDO polynomial. We can specify this result as another necessary and sufficient condition for a GDO monomial to be a GAPN function using companion matrices.

Corollary 3. *Let $F(x) = x^d$ a GDO monomial function where $d = \sum_{i=0}^{\ell} k_i.p^i$ and $k_\ell \neq 0$. Let*

$$C := \begin{bmatrix} 0 & 0 & \dots & 0 & -\frac{k_0}{k_\ell} \\ 1 & 0 & \dots & 0 & -\frac{k_1}{k_\ell} \\ \vdots & \vdots & \ddots & 0 & \vdots \\ 0 & 0 & \dots & 1 & -\frac{k_{\ell-1}}{k_\ell} \end{bmatrix} \in \mathbb{F}_p^{\ell \times \ell}.$$

Then, the function F is GAPN over $\mathbb{F}_{p^n}$ if, and only if, $\dim_{\mathbb{F}_p}(\ker(C^n - I_\ell)) = 1$.

Proof. For the monomial case, we have already seen that a GDO function F is GAPN if, and only if, $\nabla_1 F(x) = 0$ has exactly p solutions. As $\nabla_1 F(x) = -\sum_{i=0}^{\ell} k_i x^{p^i} \in \mathbb{F}_p[x]$, Applying Theorem 21 concludes the proof. □

Similarly, we can use the notations previously introduced to re-interpret the following results from McGuire and Sheekey [21].

Lemma 22 ([21] Theorem 7)**.** *Let $L(x) = \sum_{i=0}^{\ell} a_i.x^{p^i}$ an $\mathbb{F}_p$-function over $\mathbb{F}_{p^n}$ where $a_\ell \neq 0$ and $a_0 \neq 0$. L has p^ℓ roots in $\mathbb{F}_{p^n}$ if, and only if, $C^{(1)}(L) \times C^{(p)}(L) \times \cdots \times C^{(p^{n-1})}(L) = I_\ell$ where:*

$$C^{(p^i)}(L) = \begin{bmatrix} 0 & 0 & \dots & 0 & -\left(\frac{a_0}{a_\ell}\right)^{p^i} \\ 1 & 0 & \dots & 0 & -\left(\frac{a_1}{a_\ell}\right)^{p^i} \\ \vdots & \vdots & \ddots & 0 & \vdots \\ 0 & 0 & \dots & 1 & -\left(\frac{a_{\ell-1}}{a_\ell}\right)^{p^i} \end{bmatrix} \in \mathbb{F}_{p^n}^{\ell \times \ell}.$$

Using the above result, we can now provide a necessary and sufficient condition for a certain type of multinomial GDO functions with low Hamming degree (i.e., 3) and relatively low polynomial degree, to be GAPN functions.

Theorem 23. *Let p be an odd prime number, and let $F(x) = \sum_{i=1}^{N} f_i.x^{d_i} \in \mathbb{F}_{p^n}[x]$ be a GDO polynomial where the exponents d_i are of the form:*

$$d_i = d_{i,0} + d_{i,1}.p + d_{i,2}.p^2,$$

and such that the polynomials $g_0^{(F)}, g_1^{(F)}$ and $g_2^{(F)}$ have no common roots in $\mathbb{F}_{p^n}^*$.
For any $\alpha \in \mathbb{F}_{p^n}^*$ such that $g_2^{(F)}(\alpha) \neq 0$, we have

$$C^{(p^i)}(\nabla_\alpha F) = \begin{bmatrix} 0 & -\left(\frac{g_0^{(F)}(\alpha)}{g_2^{(F)}(\alpha)}\right)^{p^i} \\ 1 & -\left(\frac{g_1^{(F)}(\alpha)}{g_2^{(F)}(\alpha)}\right)^{p^i} \end{bmatrix}.$$

Then, F is a GAPN function over $\mathbb{F}_{p^n}$ if, and only if,

$$C^{(1)}(\nabla_\alpha F) \times C^{(p)}(\nabla_\alpha F) \times \cdots \times C^{(p^{n-1})}(\nabla_\alpha F) \neq I_2$$

for all such suitable α.

Proof. Let $\alpha \in \mathbb{F}_{p^n}^*$.

If $g_i^{(F)}(\alpha) = 0$ for every $i \in \{0, 1, 2\}$, then, from the formula in Proposition 16, $\nabla_\alpha F(x) = 0$ for every $x \in \mathbb{F}_{p^n}$. Then F is not a GAPN function over $\mathbb{F}_{p^n}$.

Otherwise, we have two cases depending on the value of $g_2^{(F)}(\alpha)$.

1. If $g_2^{(F)}(\alpha) = 0$, then, $\nabla_\alpha F(x)$ is a function of polynomial degree $\deg_P(\nabla_\alpha F) \leq p$, that is, it has at most p roots.

2. If $g_2^{(F)}(\alpha) \neq 0$, then applying the result from Lemma 22, $\nabla_\alpha F(x)$ has strictly less than p^2 roots in $\mathbb{F}_{p^n}$ if, and only if,

$$C^{(1)}(\nabla_\alpha F) \times C^{(p)}(\nabla_\alpha F) \times \cdots \times C^{(p^{n-1})}(\nabla_\alpha F) \neq I_2. \tag{1}$$

   Moreover, since F is a GDO function, $\nabla_\alpha F(\alpha) = 0$ always holds. So, $\nabla_\alpha F(x)$ has exactly p roots in $\mathbb{F}_{p^n}$ if, and only if, Inequality (1) is satisfied.

□

From Theorem 23, we can extract a sufficient condition for such functions to be GAPN. To do so, we use the fact that, if a determinant of some matrix $A \in \mathbb{F}_q^{n\times n}$ is not equal to 1, then A is not the identity matrix.

Corollary 4. *Let p be an odd prime number, $n \geq 3$, and let $F(x) = \sum_{i=1}^{N} f_i.x^{d_i} \in \mathbb{F}_{p^n}[x]$ be a GDO polynomial where $d_i = \sum_{j=0}^{2} d_{i,j}p^j$.*
Suppose that the equation:

$$N_n(g_2^{(F)}(X)) - N_n(g_0^{(F)}(X)) = 0$$

has no solutions in $\mathbb{F}_{p^n}^*$, where $N_n(x) = x^{\frac{p^n-1}{p-1}}$ is the absolute norm. Then, F is a GAPN function over $\mathbb{F}_{p^n}$.

Proof. If defined, the determinant of $C^{(p^i)}(\nabla_\alpha F)$ is $\left(g_0^{(F)}(\alpha)/g_2^{(F)}(\alpha)\right)^{p^i}$. Thus,

$$det\left(C^{(1)}(\nabla_\alpha F) \times C^{(p)}(\nabla_\alpha F) \times \cdots \times C^{(p^{n-1})}(\nabla_\alpha F)\right) = N_n\left(\frac{g_0^{(F)}(\alpha)}{g_2^{(F)}(\alpha)}\right).$$

Now, if $N_n(g_2^{(F)}(\alpha)) - N_n(g_0^{(F)}(\alpha)) = 0$, then

$$det\left(C^{(1)}(\nabla_\alpha F) \times C^{(p)}(\nabla_\alpha F) \times \cdots \times C^{(p^{n-1})}(\nabla_\alpha F)\right) \neq 1,$$

and this product of matrices cannot be equal to the identity matrix. Applying Theorem 23 concludes the proof. □

We use this last corollary to give the following criteria about GDO polynomials with only one exponent of Hamming degree 3.

Theorem 24. *Let p be an odd prime number, $n \geq 3$ be an integer, and let $F(x) \in \mathbb{F}_{p^n}[x]$ be the GDO polynomial*

$$F(x) = x^d + \sum_{i\in\mathbb{F}_p^*} \lambda_i.x^{k_i},$$

where $d = d_2p^2 + (p - (d_0 + d_2))p + d_0$, $d_2 \neq 0$ and $k_i = (p - i)p + i$.
If

$$N_n(\frac{d_0}{d_2} + \sum_{i\in\mathbb{F}_p^*} \lambda_i.\frac{i}{d_2}\alpha^{k_i-d}) \neq 1 \quad \forall\, \alpha \in \mathbb{F}_{p^n}^*,$$

then F is a GAPN function over $\mathbb{F}_{p^n}$.

Proof. Using Lemma 6, we can compute $g_0^{(F)}(\alpha)$ and $g_2^{(F)}(\alpha)$. We get:

$$g_0^{(F)}(\alpha) = -d_0\alpha^{d-1} - \sum_{i\in\mathbb{F}_p^*} i.\lambda_i\alpha^{k_i-1},$$

and

$$g_2^{(F)}(\alpha) = -d_2\alpha^{d-p^2}.$$

As $g_2^{(F)}(\alpha)$ is a monomial in α, it has only one root in $\mathbb{F}_{p^n}$, which is 0. Thus $g_0^{(F)}$, $g_1^{(F)}$ and $g_2^{(F)}$ have no common roots in $\mathbb{F}_{p^n}^*$.
From Corollary 4, if $N_n(g_2^{(F)}(\alpha)) - N_n(g_2^{(F)}(\alpha)) \neq 0$ for all $\alpha \in \mathbb{F}_{p^n}^*$, then F is GAPN over $\mathbb{F}_{p^n}$. We have

$$\begin{aligned}
N_n(g_2^{(F)}(\alpha)) - N_n(g_2^{(F)}(\alpha)) \neq 0 &\iff N_n(d_2\alpha^{d-p^2}) \neq N_n(d_0\alpha^{d-1} + \sum_{i\in\mathbb{F}_p^*} i.\lambda_i\alpha^{k_i-1}) \\
&\iff N_n(d_0).N_n(\alpha)^{d-p^2} \neq N_n(\alpha^{d-1})N_n(d_0 + \sum_{i\in\mathbb{F}_p^*} i.\lambda_i\alpha^{k_i-d}) \\
&\iff N_n(d_0)N_n(\alpha)^{1-1} \neq N_n(\alpha)^{d-1}N_n(d_0 + \sum_{i\in\mathbb{F}_p^*} i.\lambda_i\alpha^{k_i-d}) \\
&\iff N_n(d_0) \neq N_n(\alpha)^{1-1}N_n(d_0 + \sum_{i\in\mathbb{F}_p^*} i.\lambda_i\alpha^{k_i-d}) \\
&\iff 1 \neq N_n(\frac{d_0}{d_2} + \sum_{i\in\mathbb{F}_p^*} \frac{i}{d_2}.\lambda_i\alpha^{k_i-d}).
\end{aligned}$$

□

As we already see, we can use Theorem 20 and Lemma 22 to have many criteria to find GDO multinomials, which are combinations of GDO monomials with exponents of the form $d_2p^2+d_1p+d_0$. As a further demonstration, we deduce the following sufficient condition for GDO binomials.

Corollary 5. *Let p be an odd prime number, $n \geq 3$ be an integer, and, let*

$$F(x) = x^{(p-d_0)p^2+d_0p} + \lambda x^{k_0p+(p-k_0)} \in \mathbb{F}_{p^n}[x]$$

be a GDO binomial where $\lambda \in \mathbb{F}_{p^n}^*$. If $N_n(\lambda\frac{k_0}{d_0}) \neq 1$, then F is a GAPN function over $\mathbb{F}_{p^n}$.

Proof. Denote $d = (p-d_0)p^2 + d_0p$ and $k = k_0p + (p-k_0)$. From Lemma 6 we have that $\nabla_\alpha F(x) = d_0\alpha^{d-p^2}x^{p^2} - g_1^F(\alpha)x^p + \lambda k_0\alpha^{k-1}x$. We also have

$$\begin{aligned}
N_n(g_0^F(\alpha)) = N_n(g_2^F(\alpha)) &\iff N_n(d_0\alpha^{d-p^2}) = N_n(\lambda k_0\alpha^{k-1}) \\
&\iff N_n(\frac{k_0}{d_0}\lambda) = \frac{N_n(\alpha^{d-p^2})}{N_n(\alpha^{k-1})} \\
&\iff N_n(\frac{k_0}{d_0}\lambda) = N_n(\alpha)^{d-k} \\
&\iff N_n(\frac{k_0}{d_0}\lambda) = N_n(\alpha)^{p-d_0+d_0-(k_0+p-k_0)} \\
&\iff N_n(\frac{k_0}{d_0}\lambda) = N_n(\alpha)^0 = 1.
\end{aligned}$$

We end the proof using Corollary 4. □

Example 7. Let p be an odd prime number. The following function $F(x) = x^{(p-i)p^2+ip} + \lambda x^{ip+p-i}$ where $i \in \{1, \ldots, p-1\}$ is GAPN over $\mathbb{F}_{p^n}$ for every positive integer $n \geq 3$ such that $\lambda^{\frac{p^n-1}{p-1}} \neq 1$

where $\lambda \in \mathbb{F}_{p^n}$.

For a fixed p and n, we can always find $\lambda \in \mathbb{F}_{p^n}^*$ such that $\lambda^{\frac{p^n-1}{p-1}} \neq 1$. In fact, $\{\lambda \in \mathbb{F}_{p^n} \mid \lambda^{\frac{p^n-1}{p-1}} = 1\}$ has a cardinality at most $\frac{p^n-1}{p-1} < p^n - 1$ because p is odd.

5.2.1 GDO multinomials of Hamming degree 3

We need the following definition for our next results.

Definition 15. Let $p \geq 3$ be a prime number. For $1 \leq k \leq p-1$, we define $e_3(k)$ as

$$e_3(k) := kp^2 + (p-2k)p + k.$$

Theorem 25. *Let $p \geq 5$ be a prime number, $n \geq 3$ be an integer, and, let*

$$F(x) = \sum_{i=1}^{\frac{p-1}{2}} f_i x^{e_3(i)} \in \mathbb{F}_{p^n}[x]$$

be a GDO polynomial. Then, F is a GAPN function over $\mathbb{F}_{p^n}$ if, and only if, $n \not\equiv 0 \pmod p$ and the polynomial

$$g(X) = \sum_{i=1}^{\frac{p-1}{2}} i f_i X^{e_3(i)} \in \mathbb{F}_{p^n}[X]$$

has no root in $\mathbb{F}_{p^n}^*$.

Proof. Let $\alpha \in \mathbb{F}_{p^n}^*$. We have

$$\begin{aligned}
\nabla_\alpha F(x) &= \sum_{i=1}^{\frac{p-1}{2}} f_i(-i\alpha^{e_3(i)-p^2}x^{p^2} + 2i\alpha^{e_3(i)-p}x^p - i\alpha^{e_3(i)-1}x) \\
&= \sum_{i=1}^{\frac{p-1}{2}} (-i) \times f_i \times (\alpha^{e_3(i)-p^2}x^{p^2} - 2\alpha^{e_3(i)-p}x^p + \alpha^{e_3(i)-1}x) \\
&= \sum_{i=1}^{\frac{p-1}{2}} (-i) \times f_i \times \alpha^{e_3(i)} \times (\alpha^{-p^2}x^{p^2} - 2\alpha^{-p}x^p + \alpha^{-1}x) \\
&= -\left(\sum_{i=1}^{\frac{p-1}{2}} i f_i \alpha^{e_3(i)}\right) \times \left((\frac{x}{\alpha})^{p^2} - 2(\frac{x}{\alpha})^p + \frac{x}{\alpha}\right).
\end{aligned}$$

Thus, if $\sum_{i=1}^{\frac{p-1}{2}} i f_i \alpha^{e_3(i)} \neq 0$, then

$$\nabla_\alpha F(x) = 0 \iff (\frac{x}{\alpha})^{p^2} - 2(\frac{x}{\alpha})^p + \frac{x}{\alpha} = 0.$$

From Lemma 14, the polynomial $X^{p^2} - 2X^p + X \in \mathbb{F}_p[X]$ has only p roots in $\mathbb{F}_{p^n}$ if, and only if, $n \not\equiv 0 \pmod p$. Thus, $(\frac{X}{\alpha})^{p^2} - 2(\frac{X}{\alpha})^p + \frac{X}{\alpha}$ has only p roots in $\mathbb{F}_{p^n}$ if, and only if, $n \not\equiv 0 \pmod p$. We conclude that the equation $\nabla_\alpha F(x) = 0$ has p solutions in $\mathbb{F}_{p^n}$ if, and only if, $n \not\equiv 0 \pmod p$. $\square$

We have the following result for GDO binomials.

Corollary 6. *Let $p \geq 5$ be a prime number and $F(x) = x^{e_3(i)} - \lambda x^{e_3(j)}$ be a function over $\mathbb{F}_{p^n}$ where $1 \leq i \neq j \leq \frac{p-1}{2}$ and $\lambda \in \mathbb{F}_{p^n}^*$. Then F is GAPN over $\mathbb{F}_{p^n}$ if, and only if, $n \not\equiv 0 \pmod p$ and $\frac{i}{j}\lambda$ is not a $(e_3(j) - e_3(i))$ power in $\mathbb{F}_{p^n}$.*

Proof. From Theorem 25, if

$$i\alpha^{e_3(i)} - j\lambda\alpha^{e_3(j)} \neq 0 \qquad \forall\, \alpha \in \mathbb{F}_{p^n}^*,$$

then $F(x) = x^{e_3(i)} - \lambda x^{e_3(j)}$ is GAPN over $\mathbb{F}_{p^n}$ if, and only if, $n \not\equiv 0 \pmod p$. On the other hand, for $\alpha \in \mathbb{F}_{p^n}^*$, we have that

$$i\alpha^{e_3(i)} - j\lambda\alpha^{e_3(j)} \neq 0 \iff \frac{i}{j}\lambda \neq \alpha^{e_3(j)-e_3(i)}.$$

□

Remark 10. Notice that $e_3(j) - e_3(i) = (j-i)(p-1)^2$. Hence, if there exists $y \in \mathbb{F}_{p^n}$ such that $\frac{i}{j}\lambda = y^{e_3(j)-e_3(i)}$, then $N_n(\frac{i}{j}\lambda) = 1$.
Thus, if $N_n(\frac{i}{j}\lambda) \neq 1$, then $F(x) = x^{e_3(i)} - \lambda x^{e_3(j)}$ is a *GAPN* function over $\mathbb{F}_{p^n}$ if, and only if, $n \not\equiv 0 \pmod p$.
For a fixed i and j, we can always find $\lambda \in \mathbb{F}_{p^n}^*$ such that $N_n(\frac{i}{j}\lambda) \neq 1$, i.e., such that F is a GAPN function as in Corollary 6.

Example 8. Let $p \geq 5$ be a prime number. By Theorem 25, we conclude that the binomial

$$F(x) = x^{p^2+(p-2)p+1} - \lambda x^{2p^2+(p-4)p+2}$$

where $\frac{\lambda}{2}$ is not a $(p-1)^2$ power in $\mathbb{F}_{p^n}^*$, is GAPN over $\mathbb{F}_{p^n}$ for every $n \geq 3$ such that $n \not\equiv 0 \pmod p$. By Remark 10, the function

$$F(x) = x^{p^2+(p-2)p+1} + 2x^{2p^2+(p-4)p+2}$$

is GAPN over $\mathbb{F}_{p^n}$ for every odd integer $n \geq 3$ such that $n \not\equiv 0 \pmod p$. It is a new exceptional Hamming degree 3 binomial GAPN function.

As $e_3(i) - e_3(j) = (i-j)(p-1)^2$ for all $1 \leq i, j \leq \frac{p-1}{2}$, we can use Theorem 25 to derive the following result.

Corollary 7. *Let $p \geq 5$ be a prime number and $F(x) = \sum_{i=1}^{\frac{p-1}{2}} f_i x^{e_3(i)} \in \mathbb{F}_{p^n}[x]$ be a GDO polynomial, where $n \geq 3$. Suppose that $f_1 \neq 0$ and that the polynomial $g(X) = \sum_{i=1}^{\frac{p-1}{2}} i f_i X^{i-1} \in \mathbb{F}_{p^n}[X]$ has no roots in $\mathbb{F}_{p^n}^*$, then F is a GAPN function over $\mathbb{F}_{p^n}$ if, and only if, $n \not\equiv 0 \pmod p$.*

Proof. From Theorem 25, if $G(X) = \sum_{i=1}^{\frac{p-1}{2}} i f_i X^{e_3(i)} \in \mathbb{F}_{p^n}[X]$ has no roots in $\mathbb{F}_{p^n}^*$, then F is GAPN

over $\mathbb{F}_{p^n}$ if, and only if, $n \not\equiv 0 \pmod p$. On the other hand, we have:

$$\begin{aligned} G(X) &:= \sum_{i=1}^{\frac{p-1}{2}} i f_i X^{e_3(i)} \\ &= X^{e_1} \times \sum_{i=1}^{\frac{p-1}{2}} i f_i X^{e_3(i)-e_3(1)} \\ &= X^{e_3(1)} \times \sum_{i=1}^{\frac{p-1}{2}} i f_i X^{(i-1)(p-1)^2} \\ &= X^{e_3(1)} \times g(X^{(p-1)^2}). \end{aligned}$$

As a consequence, $G(X)$ and $g(X^{(p-1)^2})$ share the same nonzero roots.
In particular, if the polynomial $g(Y) = \sum_{i=1}^{\frac{p-1}{2}} i f_i Y^{(i-1)} \in \mathbb{F}_{p^n}[Y]$ has no nonzero roots in $\mathbb{F}_{p^n}$, then the same goes for $G(X)$. □

Example 9. Let $p \geq 7$ be a prime number, $n \geq 3$ be an integer such that $n \not\equiv 0 \pmod p$ and $F(x) = f_1 x^{e_3(1)} + \frac{f_2}{2} x^{e_3(2)} + \frac{f_3}{3} x^{e_3(3)}$ be a function over $\mathbb{F}_{p^n}$. By Corollary 7, if the polynomial

$$P(X) = f_3 X^2 + f_2 X + f_1 \in \mathbb{F}_{p^n}[X]$$

has no roots in $\mathbb{F}_{p^n}^*$, then F is GAPN over $\mathbb{F}_{p^n}$.
Suppose that $f_1 \neq 0$, i.e., $P(0) \neq 0$. Then P has roots in $\mathbb{F}_{p^n}^*$ if, and only if, $f_2^2 - 4f_1 f_3$ is not a square in $\mathbb{F}_{p^n}$.
As a consequence, if $f_2^2 - 4f_1 f_3$ is not a square in $\mathbb{F}_{p^n}$, then F is a GAPN function over $\mathbb{F}_{p^n}$.

In Example 9, we can always suppose that $f_1 = 1$ (GEA-equivalence), and if $f_2^2 - 4f_3$ is not a square in $\mathbb{F}_{p^n}$, then the function F is GAPN. As a consequence, we have the following corollary.

Corollary 8. *Let $p \geq 7$ be a prime number, $n \geq 3$ is such that $n \not\equiv 0 \pmod p$ and $\lambda \in \mathbb{F}_{p^n}$ be a non square. Then*

$$F(x) = x^{e_3(1)} + \frac{f}{2} x^{e_3(2)} + \frac{f^2 - \lambda}{12} x^{e_3(3)}$$

is GAPN over $\mathbb{F}_{p^n}$ for all $f \in \mathbb{F}_{p^n}^*$.

Proof. As we mentioned above, if $\lambda := f_2^2 - 4f_3 \in \mathbb{F}_{p^n}$ is not a square, then F is GAPN over $\mathbb{F}_{p^n}$. Thus, given a non square λ, we have $f_3 = \frac{f_2^2 - \lambda}{4}$. □

5.2.2 GDO multinomials of Hamming degree 5

We can extend this construction for a Hamming degree equal to 5. First, let us define the following notation.

Definition 16. Let $p \geq 5$ be a prime number. For $1 \leq k \leq p-1$ such that

- $1 \leq p - 4k \leq p - 1$, and,

- $1 \leq 6k - p \leq p - 1$,

we define $e_5(k)$ as

$$e_5(k) := kp^4 + (p - 4k)p^3 + (6k - p)p^2 + (p - 4k)p + k.$$

For a given prime $p \geq 5$, we can show that k in Definition 16 satisfy $\frac{p+1}{6} \leq k \leq \frac{p-1}{4}$. As we are interested in multinomials, it is crucial to know how many $k \in [1, p-1]$ satisfy $\frac{p+1}{6} \leq k \leq \frac{p-1}{4}$ when p is given.
This quantity is exactly:

$$I(p) := \lfloor \frac{p-1}{4} \rfloor - \lceil \frac{p+1}{6} \rceil + 1.$$

Example 10. We express $I(p)$ for different prime p:

1. For $p = 17$:

$$\begin{aligned} I(p) &= \lfloor \frac{p-1}{4} \rfloor - \lceil \frac{p+1}{6} \rceil + 1 = 2 \\ &= 4 - 3 + 1 \\ &= 2. \end{aligned}$$

2. For $p = 19$: we find $I(p) = 4 - 4 + 1 = 1$.

3. For $p = 257$: we find $I(p) = 64 - 43 + 1 = 22$.

Remark 11. For $p \geq 17$ and $p \neq 19$, we have $I(p) \geq 2$.

By a simple re-adaptation of the proof of Theorem 25, we have the following result.

Theorem 26. *Let $p \geq 17$ be a prime number such that $p \neq 19$, and,*

$$F(x) = \sum_{i=\lceil \frac{p+1}{6} \rceil}^{\lfloor \frac{p-1}{4} \rfloor} f_i x^{e_5(i)} \in \mathbb{F}_{p^n}[x]$$

be a function over $\mathbb{F}_{p^n}$ where $n \geq 5$. Then, F is GAPN over $\mathbb{F}_{p^n}$ if, and only if, $n \not\equiv 0 \pmod p$ and the following polynomial has no root in $\mathbb{F}_{p^n}^*$:

$$\sum_{i=\lceil \frac{p+1}{6} \rceil}^{\lfloor \frac{p-1}{4} \rfloor} i f_i X^{e_5(i)}.$$

Similarly, we have the following result on binomials as in Corollary 6.

Corollary 9. *Let $p \geq 17$ be a prime number such that $p \neq 19$. Let $F(x) = x^{e_5(i)} - \lambda x^{e_5(j)}$ be a function over $\mathbb{F}_{p^n}$ where $\lceil \frac{p+1}{6} \rceil \leq i \neq j \leq \lfloor \frac{p-1}{4} \rfloor$ and $\lambda \in \mathbb{F}_{p^n}$ is such that $\frac{i}{j}\lambda$ is not a $e_5(i) - e_5(j)$ power. Then, F is a GAPN function over $\mathbb{F}_{p^n}$ if, and only if, $n \not\equiv 0 \pmod p$.*

Remark 12. Here we need $I(p) \geq 2$ (binomial), thus, we suppose $p \geq 17$ and $p \neq 19$.

Remark 13. Remark that $e_5(i) - e_5(j) = (i-j)(p-1)^4$. If $\frac{i}{j}\lambda$ is an $e_5(i) - e_5(j)$ power, then $N_n(\frac{i}{j}\lambda) = 1$. As a consequence, if we suppose that $N_n(\frac{i}{j}\lambda) \neq 1$, then $F(x) = x^{e_5(i)} - \lambda x^{e_5(j)}$ is a GAPN function over $\mathbb{F}_{p^n}$ if, and only if, $n \not\equiv 0 \pmod p$.
For a fixed j and i, we can always find $\lambda \in \mathbb{F}_{p^n}^*$ such that $N_n(\frac{i}{j}\lambda) \neq 1$.

Example 11. Let $p = 29$ and $n = 5$. We have $n \not\equiv 0 \pmod p$, $\lceil \frac{p+1}{6} \rceil = 5$ and $\lfloor \frac{p-1}{4} \rfloor = 7$.
We take:

- $i = 5$: $e_5(i) = 5 \times 29^4 + 9 \times 29^3 + 29^2 + 9 \times 29 + 5$,
- $j = 6$: $e_5(j) = 6 \times 29^4 + 5 \times 29^3 + 7 \times 29^2 + 5 \times 29 + 6$,
- $k = 7$: $e_5(k) = 7 \times 29^4 + 1 \times 29^3 + 13 \times 29^2 + 1 \times 29 + 7$.

We can now look for f_5, f_6 and f_7 in $\mathbb{F}_{p^5}$ such that the polynomial

$$5f_5X^{e_5(5)} + 6f_6X^{e_5(6)} + 7f_7X^{e_5(7)} \in \mathbb{F}_{p^n}[X]$$

has no roots in $\mathbb{F}_{p^n}^*$.
A concrete example of such GAPN functions is found by taking $f_5 = 16$, $f_6 = 25$ and $f_7 = 13$:

$$F(x) = 16x^{e_5(5)} + 25x^{e_5(6)} + 13x^{e_5(7)}.$$

Similarly to Corollary 7, as $e_5(i) - e_5(j) = (i-j)(p-1)^4$, we have the following result.

Corollary 10. *Let $p \geq 17$ be a prime number such that $p \neq 19$, and,*

$$F(x) = \sum_{i=\lceil \frac{p+1}{6} \rceil}^{\lfloor \frac{p-1}{4} \rfloor} f_i x^{e_5(i)} \in \mathbb{F}_{p^n}[x]$$

be a function over $\mathbb{F}_{p^n}$ where $n \geq 5$. Suppose that $f_{\lceil \frac{p+1}{6} \rceil} \neq 0$ and that the polynomial

$$\sum_{i=\lceil \frac{p+1}{6} \rceil}^{\lfloor \frac{p-1}{4} \rfloor} i f_i X^{i - \lceil \frac{p+1}{6} \rceil} \in \mathbb{F}_{p^n}[X]$$

has no roots in $\mathbb{F}_{p^n}^*$. Then F is a GAPN function over $\mathbb{F}_{p^n}$ if, and only if, $n \not\equiv 0 \pmod p$.

Example 12 (Trinomials)**.** Let $p \geq 29$ be a prime such that $p \neq 31$, and,

- $i = \lceil \frac{p+1}{6} \rceil$,
- $j = \lceil \frac{p+1}{6} \rceil + 1$, and,
- $k = \lceil \frac{p+1}{6} \rceil + 2$.

Let $F(x) = \frac{1}{i}x^{e_5(i)} + \frac{f_2}{j}x^{e_5(j)} + \frac{f_3}{k}x^{e_5(k)} \in \mathbb{F}_{p^n}[x]$ be a function over $\mathbb{F}_{p^n}$ and $n \geq 5$. Suppose that $f_2^2 - 4f_3$ is not a square in $\mathbb{F}_{p^n}$. Then F is a GAPN function if, and only if, $n \not\equiv 0 \pmod p$.
In particular, for any $f \in \mathbb{F}_{p^n}^*$ and any non square $\lambda \in \mathbb{F}_{p^n}$, the function

$$F(x) = \frac{1}{i}x^{e_5(i)} + \frac{f}{j}x^{e_5(j)} + \frac{f^2 - \lambda}{4k}x^{e_5(k)} \in \mathbb{F}_{p^n}[x]$$

is GAPN over $\mathbb{F}_{p^n}$ if, and only if, $n \not\equiv 0 \pmod p$.

In order to generalize the constructions in Section 5.2.1 and Section 5.2.2, for Hamming degree greater than 5, we need equivalent formulas for $e_i(k)$, $i > 5$, such that the argument of Theorems 26 and 25 can be satisfied again. One issue we encountered concerns the bound conditions about the field's characteristic.

Open problem 2. Generalize the construction of Theorems 26 and 25 for Hamming degree greater than 5. Equivalently, find the expressions of $e_i(k)$ for $i > 5$ so that $\sum_j f_j x^{e_i(j)}$ is a GAPN functions if and only if $\sum_j j f_j x^{e_i(j)} \neq 0, \ \forall x$.

6 Conclusion

In this article, we provided new constructions of GAPN functions with algebraic degree p, with both monomial and multinomial representations. We referred to homogeneous polynomial functions of algebraic degree p as Generalized Dembowski-Ostrom (GDO) functions due to the similarity with Dembowski-Ostrom (DO) polynomials, and justified this choice by illustrating the links between GDO and DO polynomials and how they could be applied to look for GAPN functions. We also provided some cryptographic properties of these functions. More precisely, we gave a lower bound on the differential uniformity of GDO functions and discussed their Walsh coefficients.

We are confident that the newly constructed GAPN functions are indeed new up to GEA-equivalence when their conditions differ from those of previously known GAPN functions. Although we have not proved the GEA-inequivalence of our GAPN multinomial functions to any power functions, we showed their GEA-inequivalence to known GAPN power functions.

Hence, one research direction would be to study the GEA-inequivalence to power functions, similarly to what has been done concerning APN functions on binary fields (see [14, 11], for instance). Using the methods developed by Salagean et al. [28] seems a good starting point.

More generally, since GDO polynomials are very similar to DO polynomials but they offer a greater diversity (due to the Hamming degree), they make for an interesting theoretical study object. For instance, GEA-equivalence problems between GDO polynomials are similar yet more challenging than EA-equivalence between DO polynomials.

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A Known GAPN functions of algebraic degree strictly greater than p

Exponents e	**Conditions**	**Hamming degree**	**Algebraic degree**	**References**
p^n-2		n	$n(p-1)-1$	[17, Proposition 3.2]
$k_2p^{l_2}+k_1p^{l_1}$	$l_1<l_2,\ 0<k_1,k_2\le p-1,$ $p\le k_1+k_2\le 2p-2\gcd(l_2-l_1,n)=1,$ $\gcd(k_1+k_2-(p-1),p^n-1)=1$	2	$\in[p,2p-3]$	[27, Theorem 2]
$kp+p-j$	$p>3,\ 1\le j<k<p,$ $gcd(k-j+1,p^n-1)=1$	2	$p+k-j$	[36, Theorem 2.4]
$tp^\ell+p-1$	$p>3,\ 1<\ell<n-1,\ gcd(\ell,n)=1,$ t odd,$1<t<p,\ gcd(t,p^n-1)=1$	2	$p+t-1$	[36, Theorem 3.1]
$tp^{n-1}-1$	$p>3,\ 0<t<p\ gcd(p-t,p^n-1)=1,\ t$ even	unknown	$n(p-1)-p+t$	[36, Theorem 4.1]
$k_2p^{l_2}+k_1p^{l_1}+p-1$	$p>3, l_1+l_2=n,$ $1<k_1+k_2\le p-1,$ $gcd(k_2p^{l_2}+k_1,p^n-1)=1,$ $gcd(l_2,n)=1$	3	$\in[p+1,2(p-1)-1]$	[30, Theorem 3.1]
$p^{2\ell}+(p-1)p^{\ell l}+p-3$	$p>3,\ gcd(\ell,n)=1,\ gcd(p^\ell+p-3,p^n-1)=1$	3	$2p-3$	[37, Theorem 3.1]

Table 3: Known monomial GAPN functions which are not GDO polynomials over $\mathbb{F}_{p^n}$ for an odd prime p up to GEA-equivalence.

Polynomial Representation	Conditions	Algebraic degree	References
$\sum_{i=u}^{p-1} c^{p-1-i} x^{ip^{l_2}+(p-1+u-i)p^{l_1}}$	$l_2 > l_1,\ gcd(l_2-l_1,n)=1,$ $1 \le u < p-1,\ gcd(u,p^n-1)=1$ $c=\alpha^v,\ \alpha$ primitive element of $\mathbb{F}_{p^n}$ $v \mid p-1$ and $v < p-1$	$p-1+u$	[27, Theorem 7]
$cx^{i_1p^{l_2}+(p-1+u-i_1)p^{l_1}} + x^{i_2p^{l_2}+(p-1+u-i_2)p^{l_1}}$	$l_2 > l_1,\ gcd(l_2-l_1,n)=1,$ $1 \le u \le i_1 < i_2 \le p-1,$ $gcd(u,p^n-1)=1$ $c=(-1)^{i_2-i_1+1}\frac{\binom{p-1-u}{i_2-u}}{\binom{p-1-u}{i_1-u}}\beta,$ $\beta \in \mathbb{F}_{p^n}$ of order not dividing $\frac{p^n-1}{i_3(p-1)}$ $i_3 = gcd(i_2-i, \frac{p^n-1}{p-1})=1$	$p-1+u$	[27, Theorem 8]
$x^{d_1}+ux^{d_2}$	$d_1,d_2 \in [1,p^2-1],\ d_2$ odd and u not a square in $\mathbb{F}_{p^2}$	$\in [p,2(p-1)]$	[2, Theorem 1]
$x^{d_1}+ux^{d_2}$	$d_1,d_2 \in [1,p^2-1],\ d_2$ even and $\exists N$ odd such that $N \mid p+1$ and $N \mid d_2-d_1$ and u not an N-th power in $\mathbb{F}_{p^2}$	$\in [p,2(p-1)]$	[2, Theorem 1]
$x^{2p-1}+ux^{kp+l}$	$u \in \mathbb{F}_{p^2}$ a primitive element, $k,l \in [0,p-1]$ with $l+k$ odd	$\in [p,2(p-1)]$	[2, Corollary 1]
$x^{kp+p-j}+x^{(k+1)p+p-j-1}$	$p>3,\ gcd(n,p-1) \in \{1,2\}$ $1 \le j < k < p-1$ $gcd(k-j+1,p^n-1)=1$	$k-j+p$	[36, Proposition 2.7]
$x^{(p-5)p^{n-1}-1}+\beta x^{(p-3)p^{n-1}-1}+5^{-1}\beta^2 x^{(p-1)p^{n-1}-1}$	$p \ge 7,\ gcd(5,p^{2n}-1)=1$	$(n-1)(p-1)+p-2$	[36, Proposition 4.3]

Table 4: Known non-monomial GAPN functions which are not GDO polynomials over $\mathbb{F}_{p^n}$ for an odd prime p.